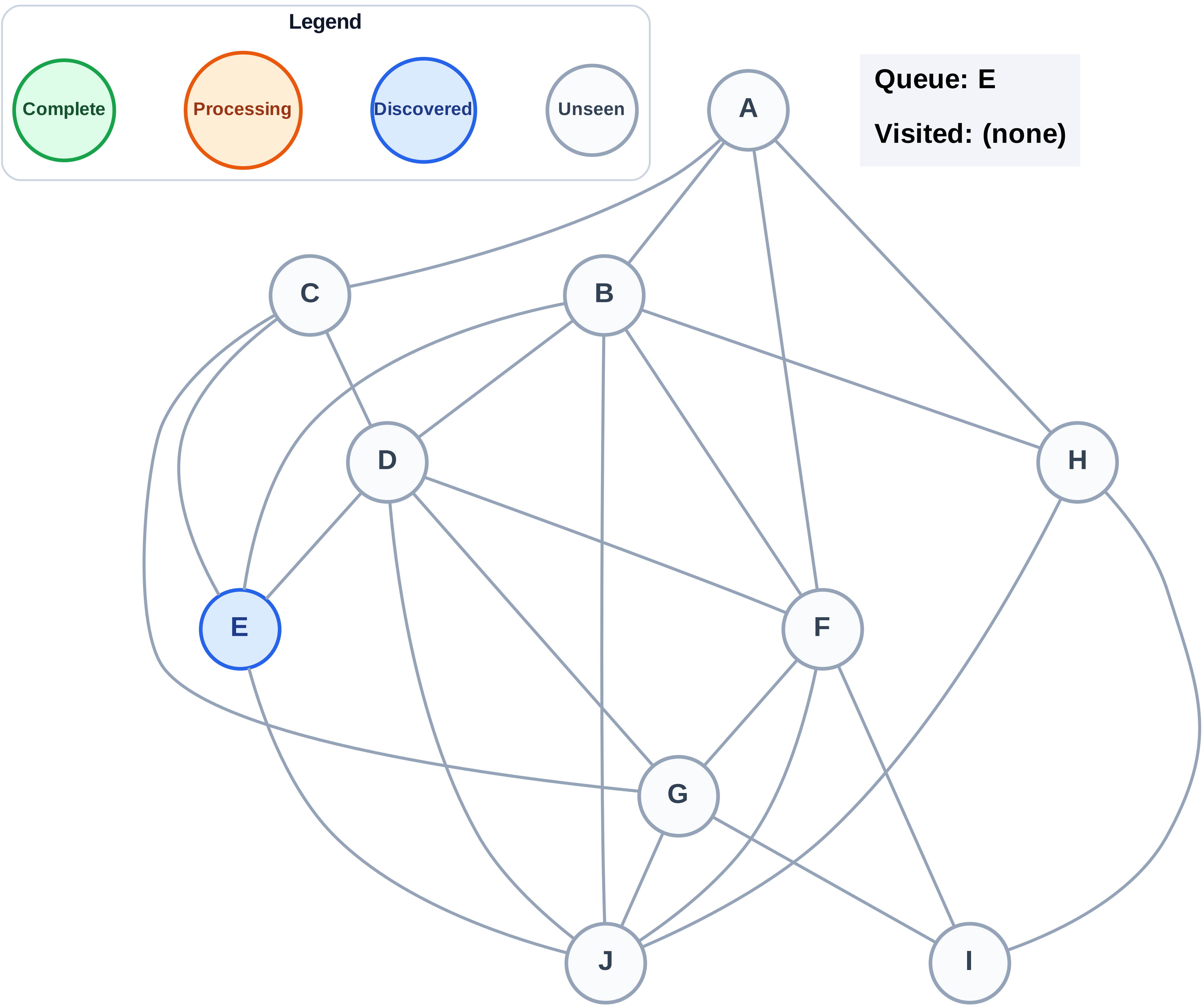
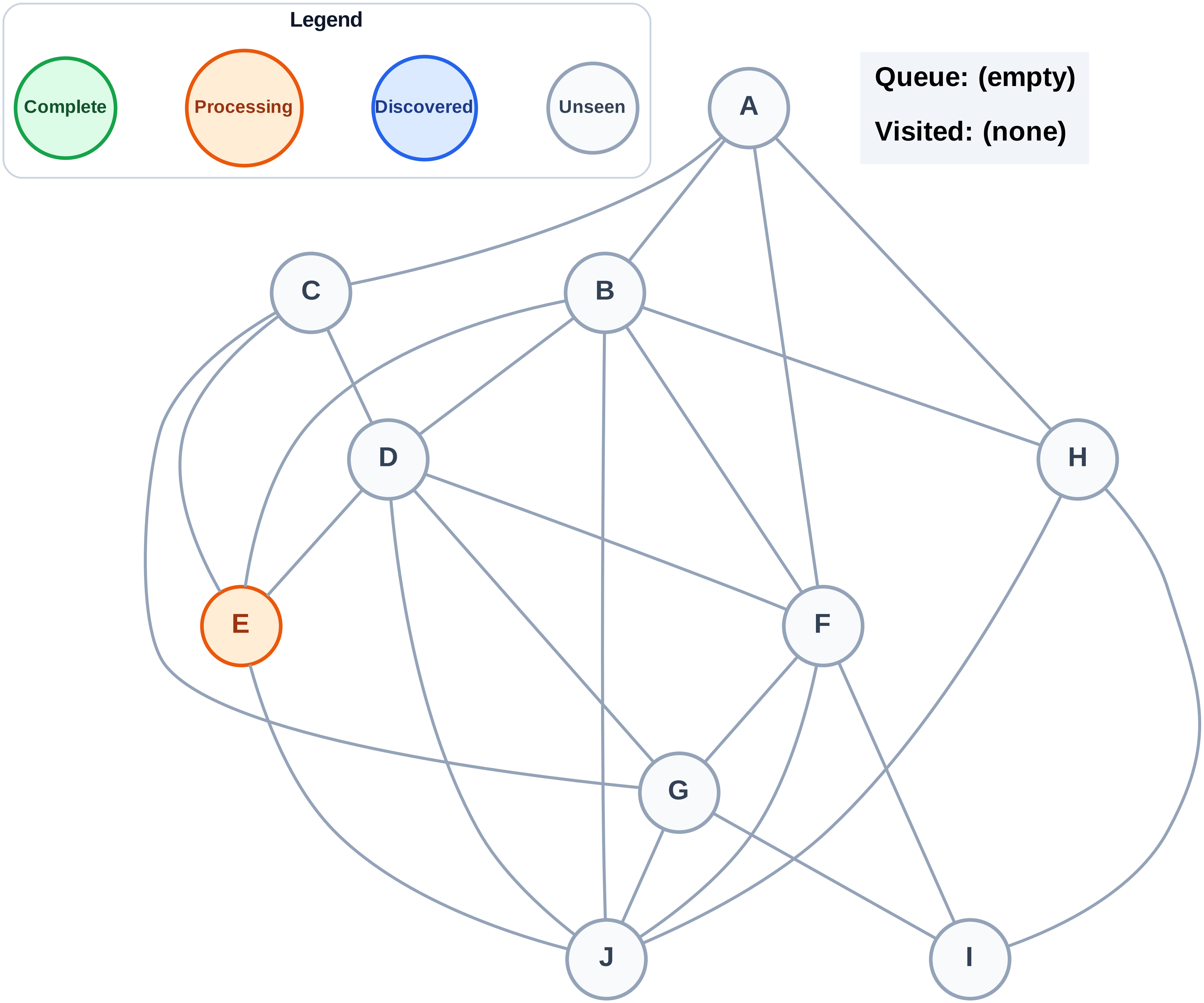
BFS Traversal

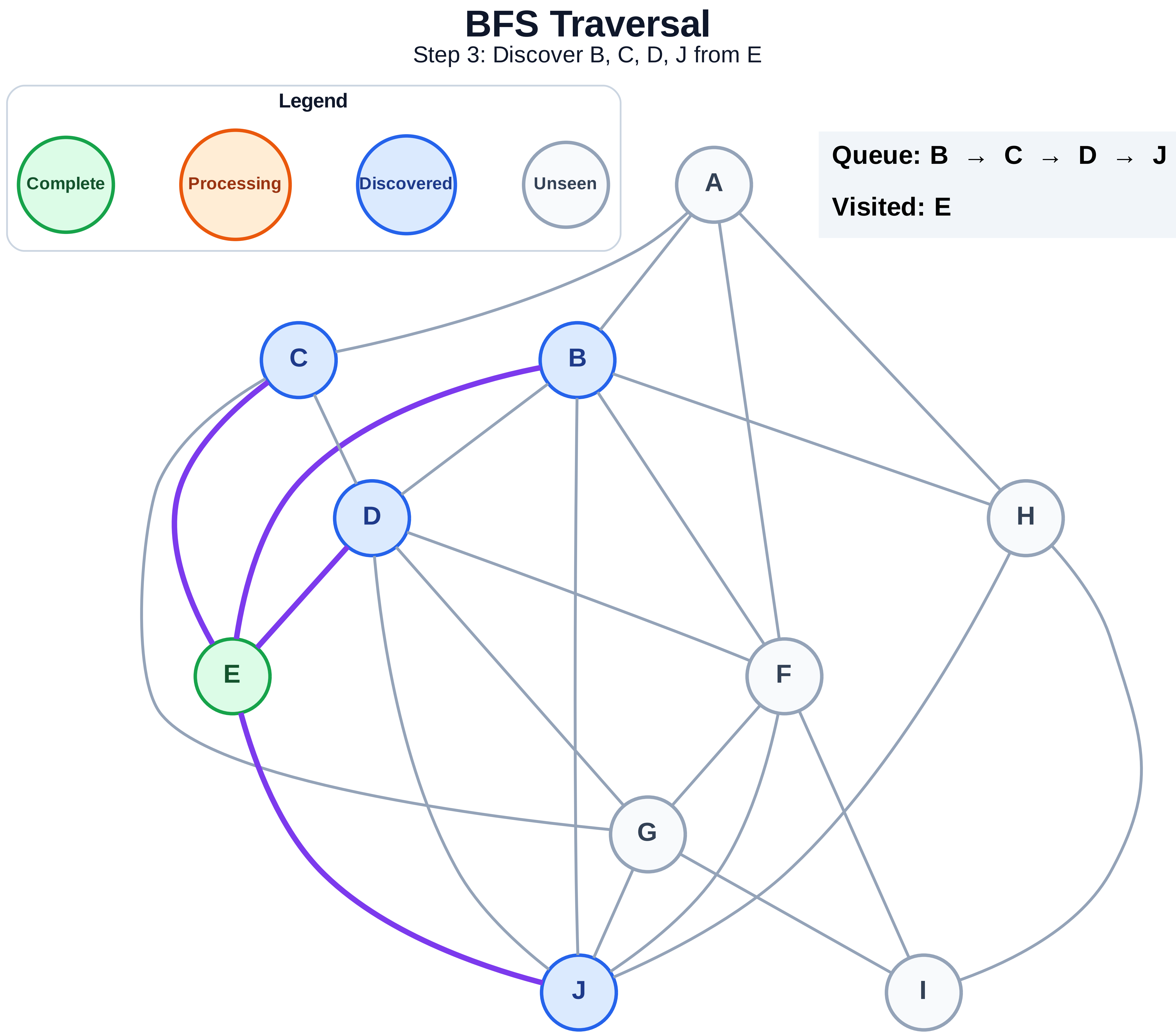
Step 1: Start at E

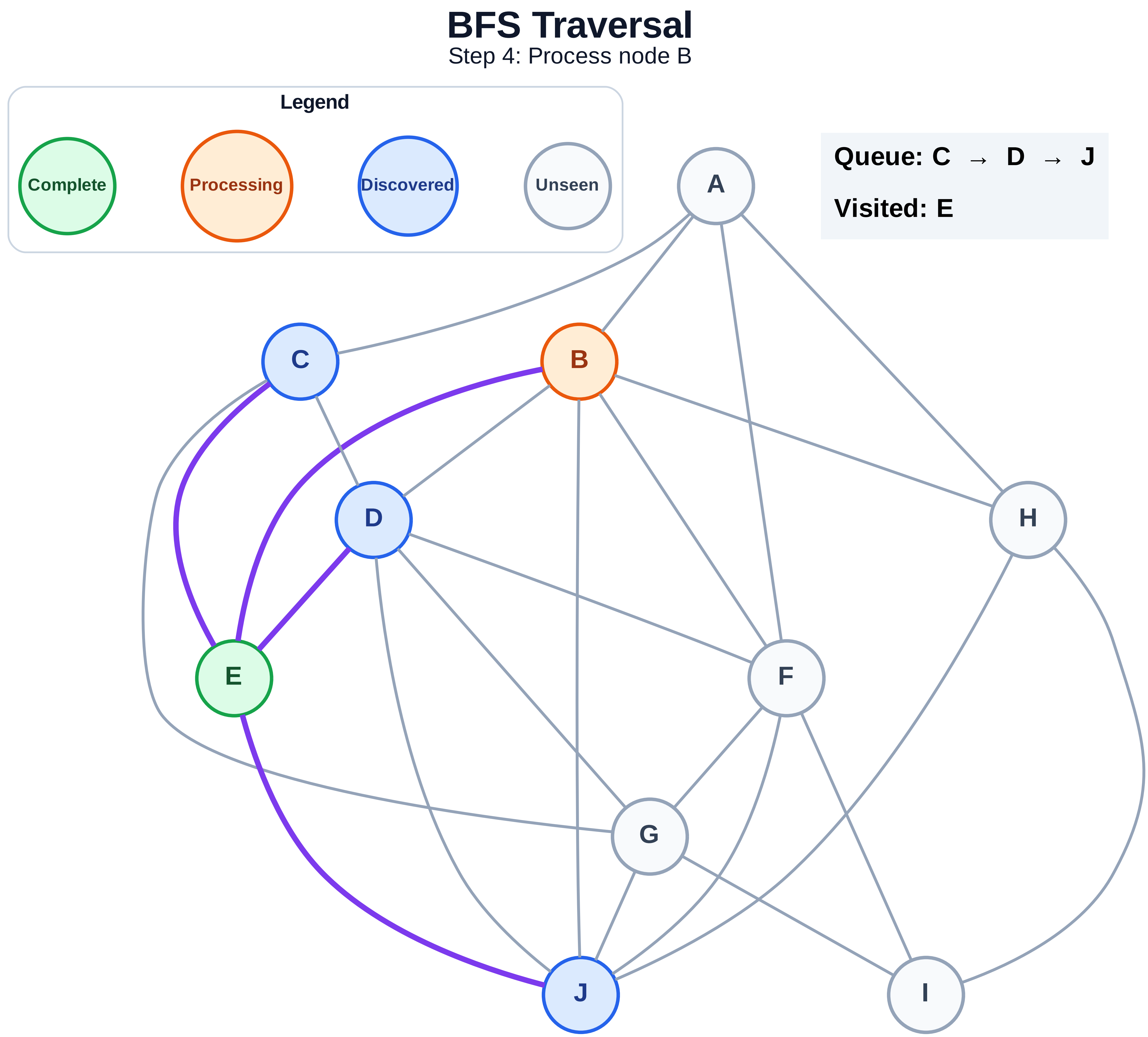


BFS Traversal

Step 2: Process node E

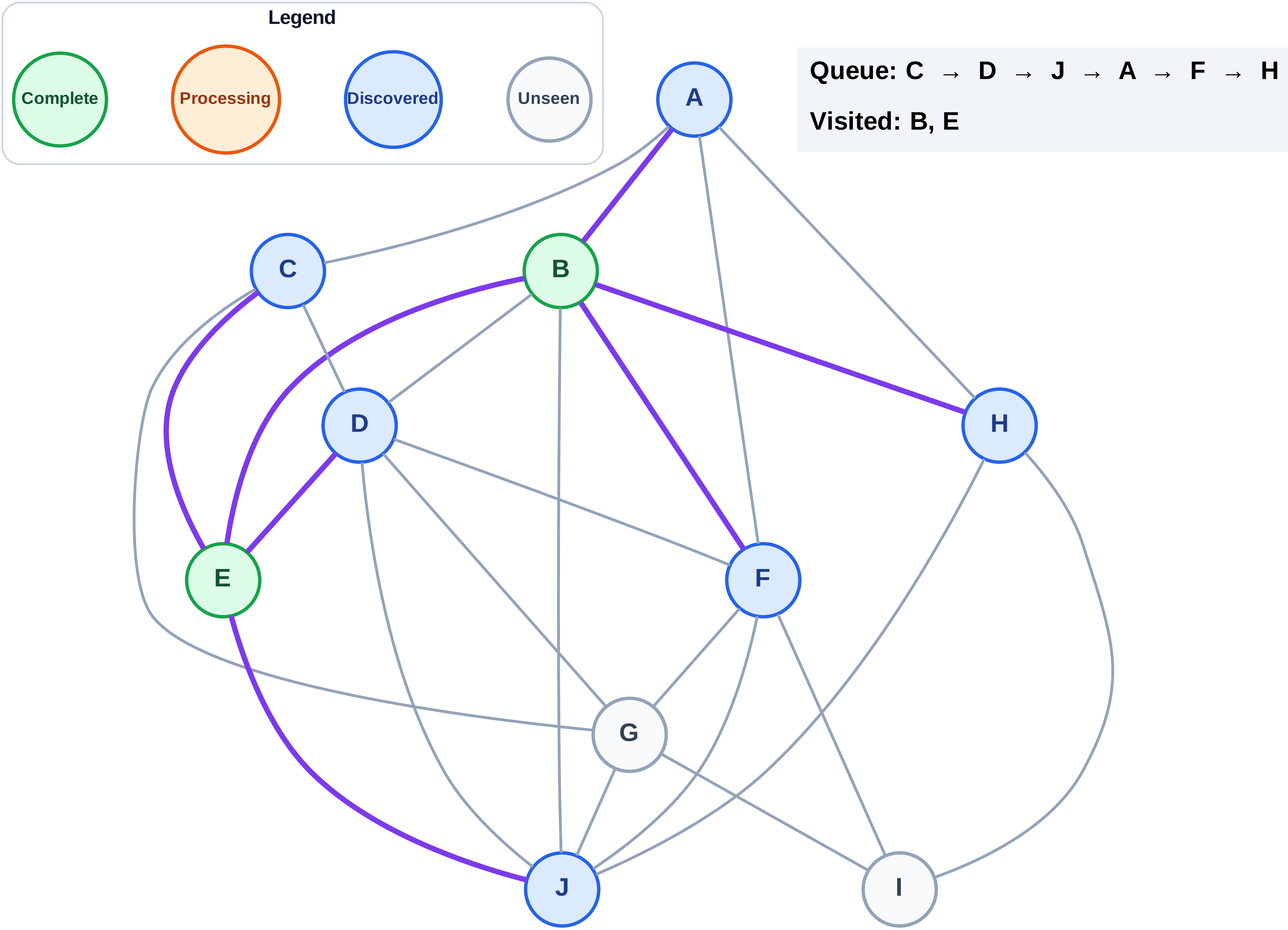



BFS Traversal
Step 3: Discover B, C, D, J from E
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → D → J
Visited: E
A
B
C
D
E
F
G
H
I
J


BFS Traversal
Step 4: Process node B
Legend
Complete
Processing
Discovered
Unseen
Queue: C → D → J
Visited: E
A
B
C
D
E
F
G
H
I
J

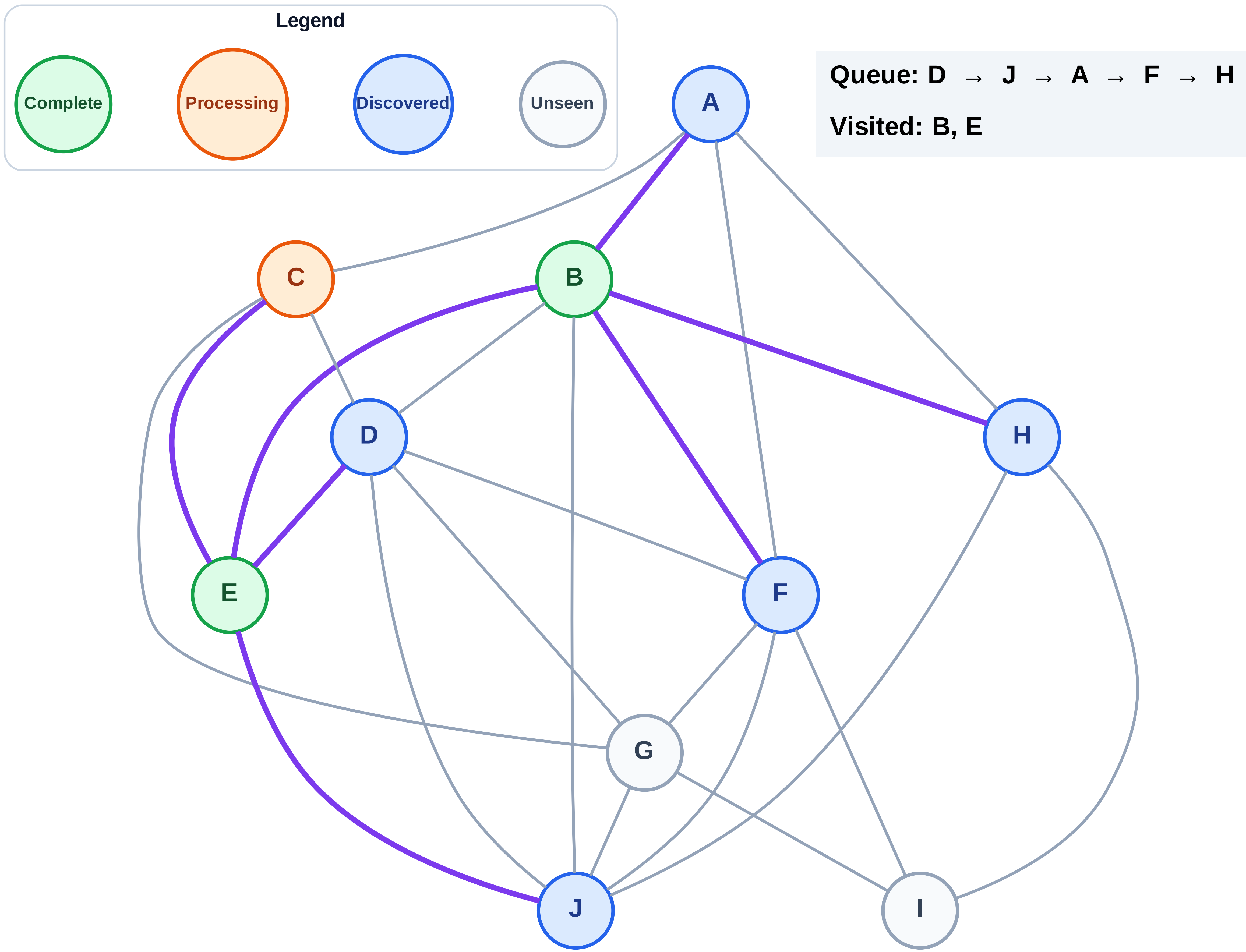
BFS Traversal

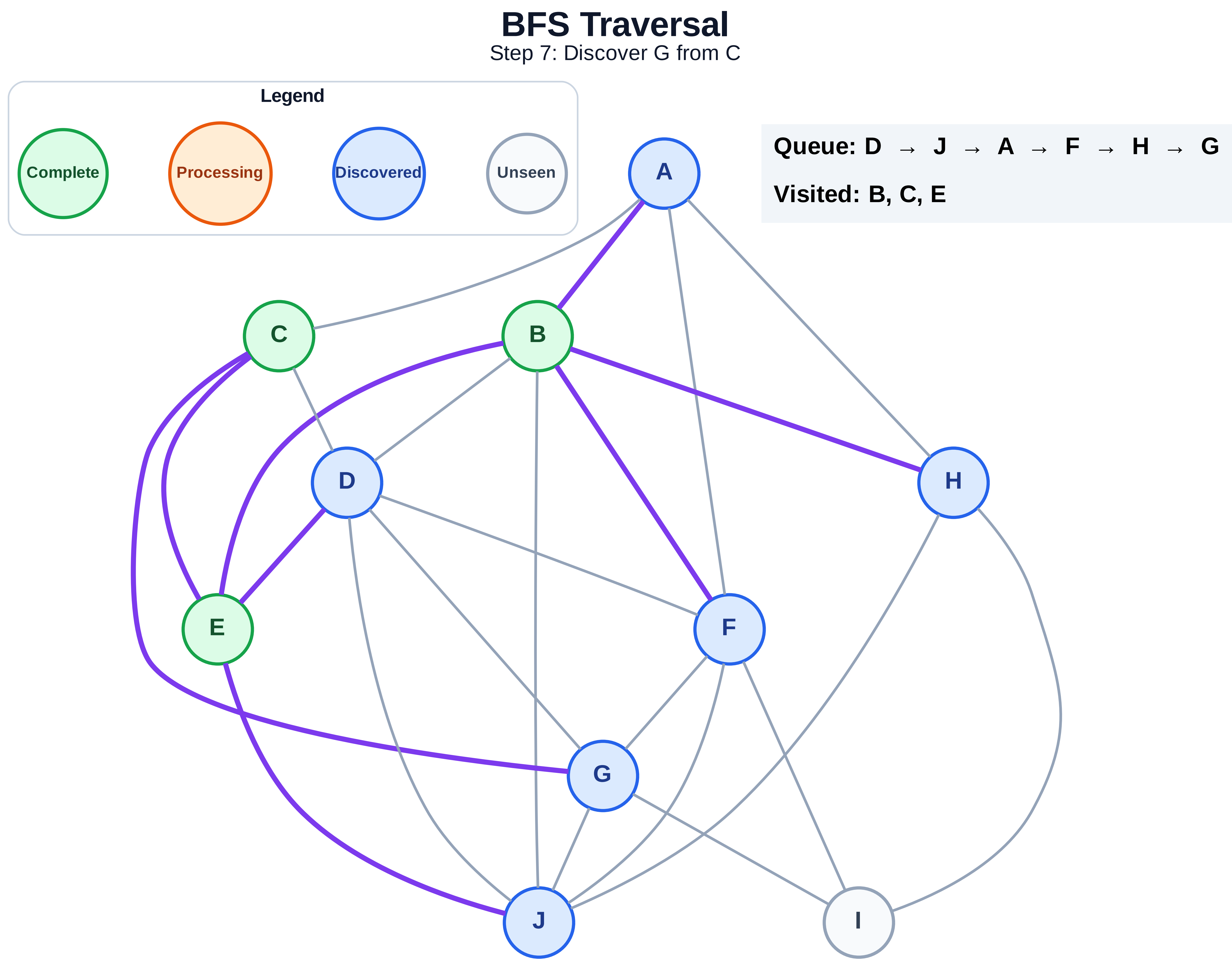
Step 5: Discover A, F, H from B



BFS Traversal

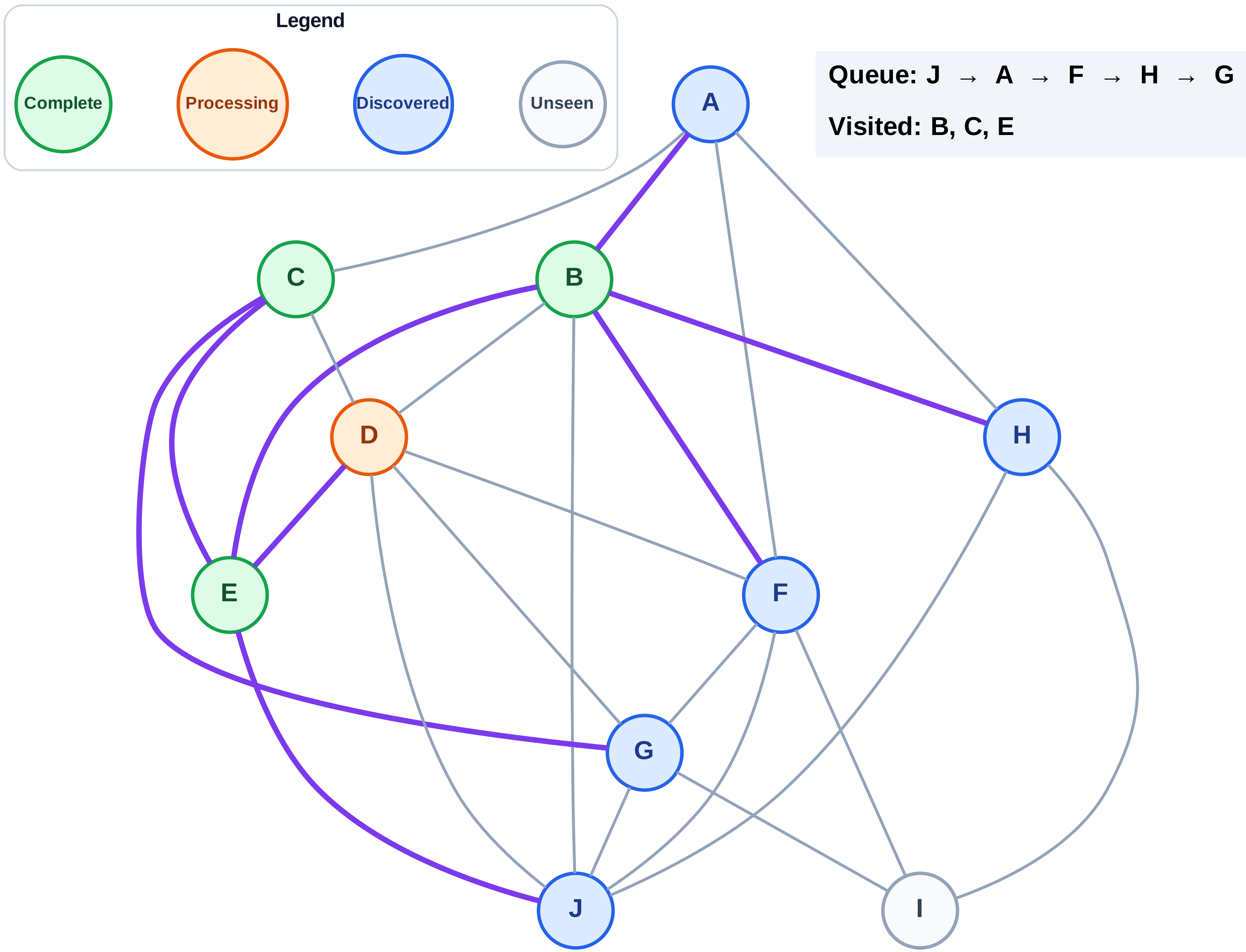
Step 6: Process node C




BFS Traversal
Step 7: Discover G from C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → J → A → F → H → G
Visited: B, C, E
A
B
C
D
E
F
G
H
I
J

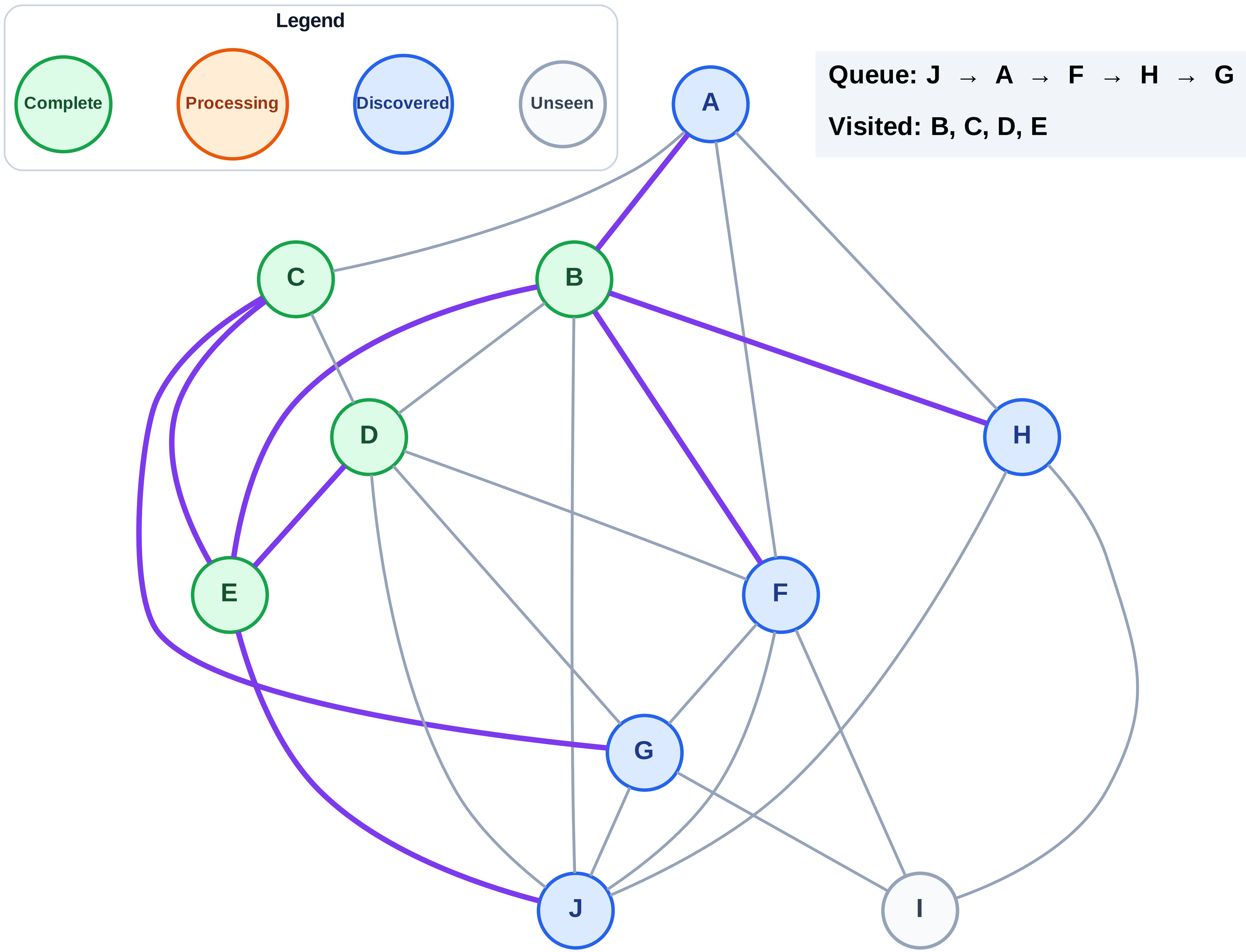
BFS Traversal

Step 8: Process node D



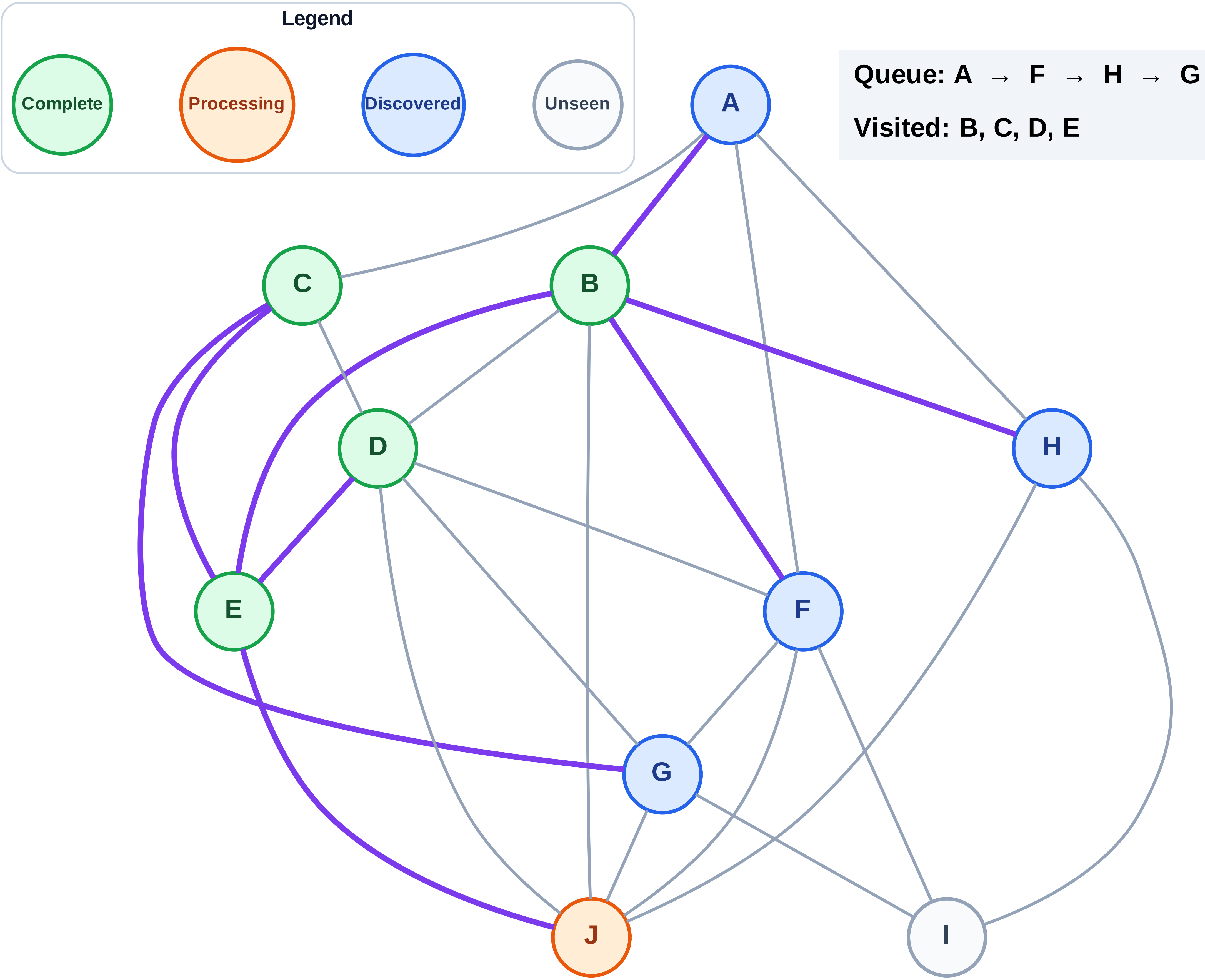
BFS Traversal

Step 9: No new neighbors from D



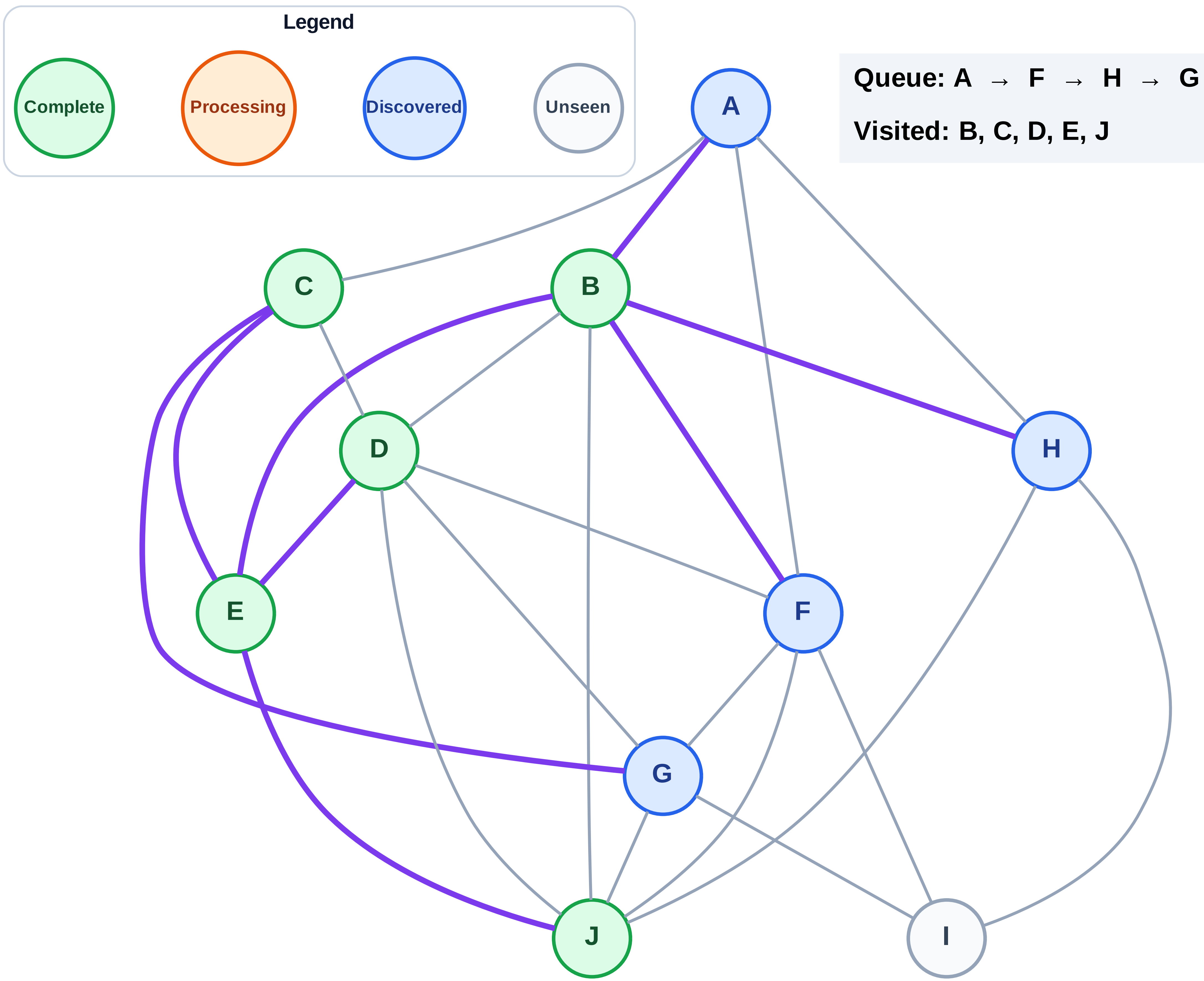
BFS Traversal

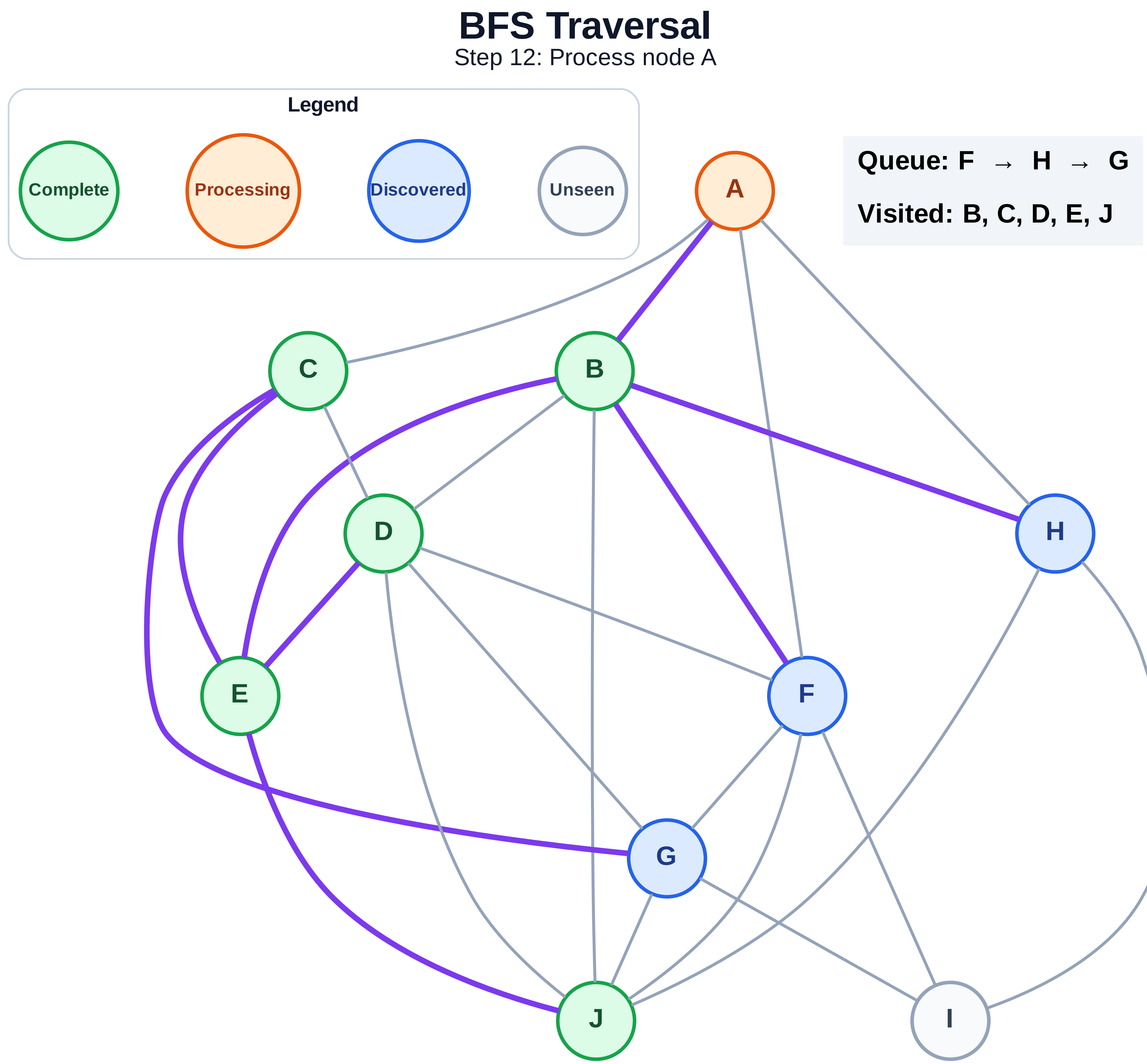
Step 10: Process node J



BFS Traversal

Step 11: No new neighbors from J

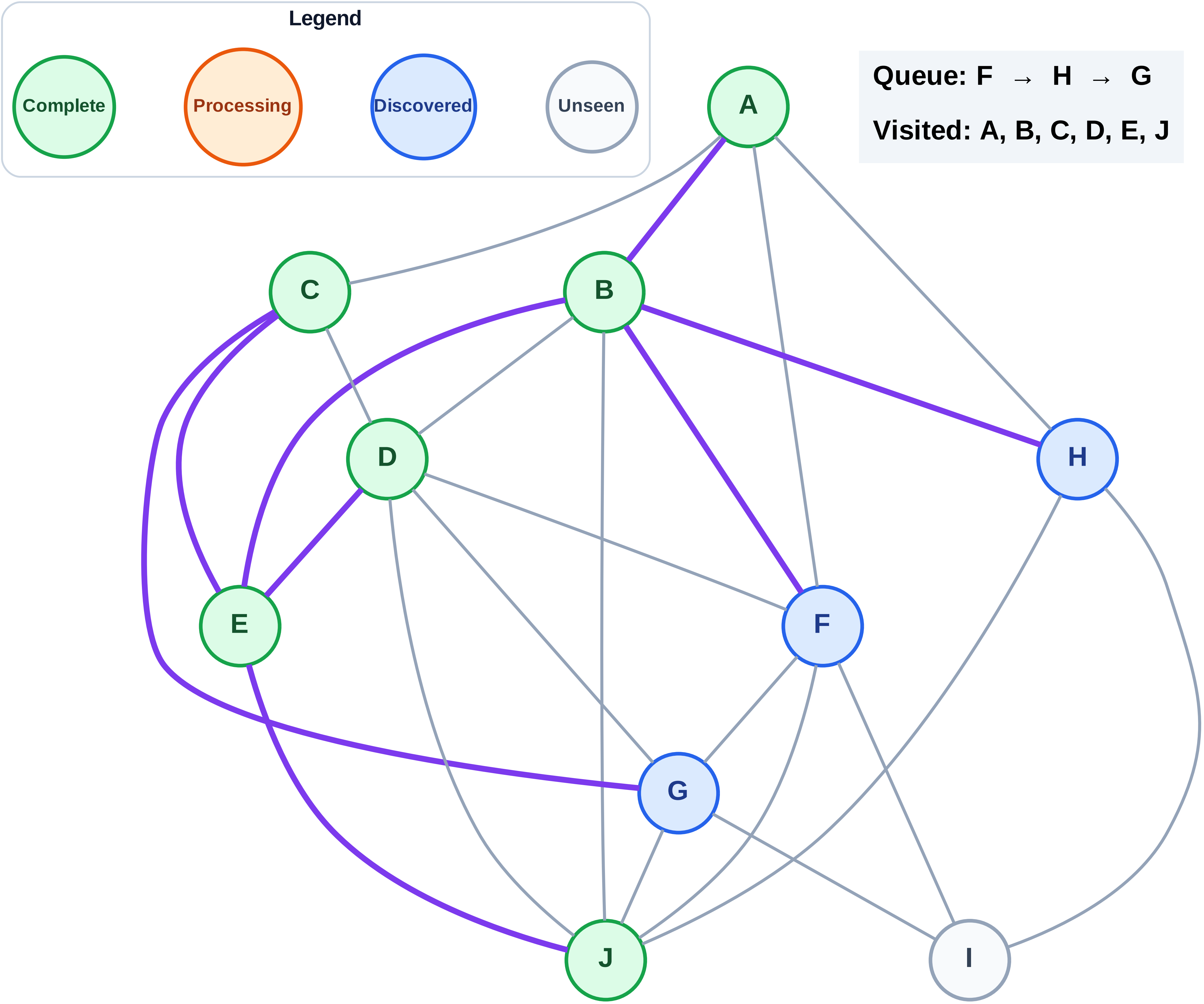




BFS Traversal
Step 12: Process node A
Legend
Complete
Processing
Discovered
Unseen
Queue: F → H → G
Visited: B, C, D, E, J
A
B
C
D
E
F
G
H
I
J

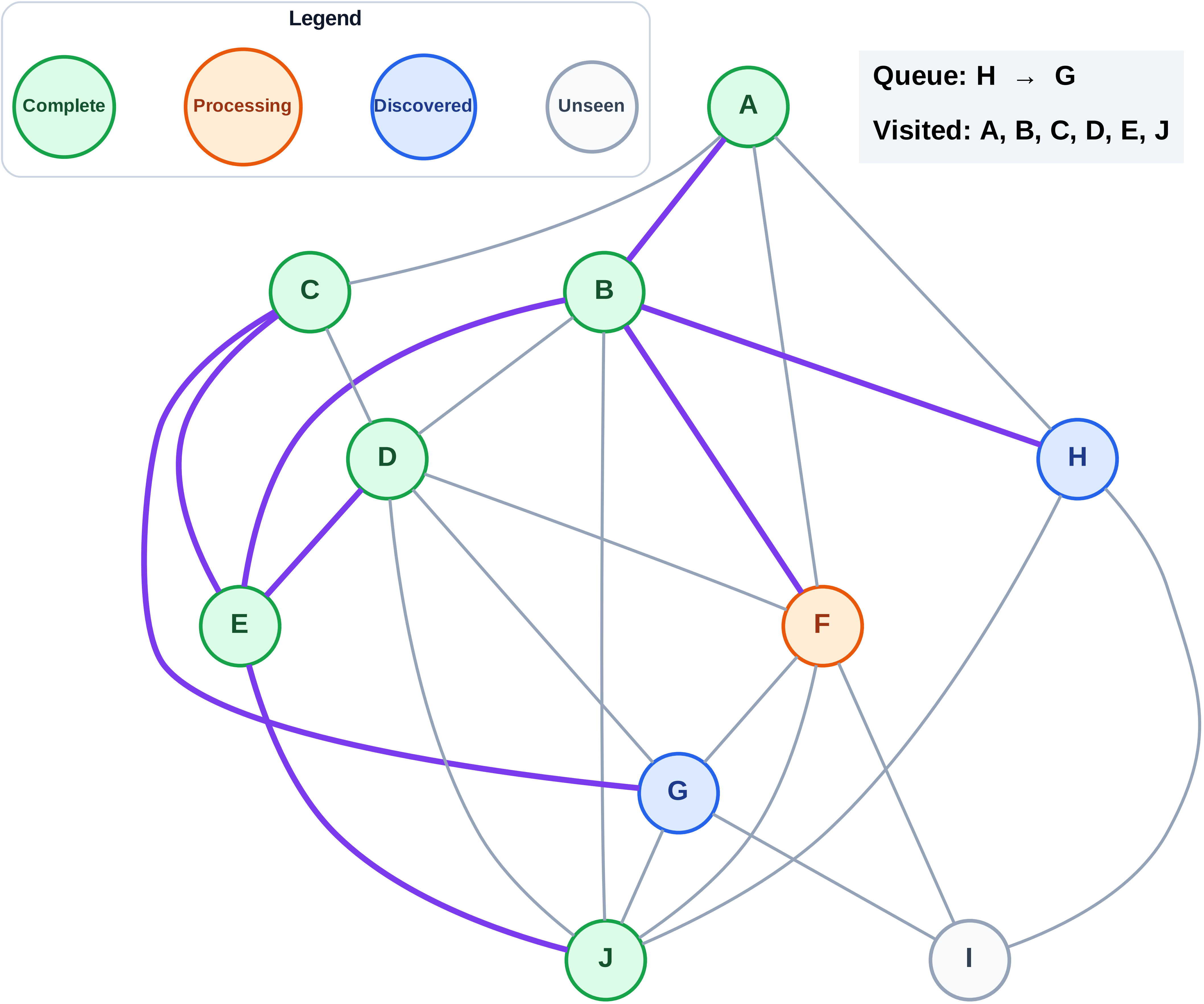
BFS Traversal

Step 13: No new neighbors from A



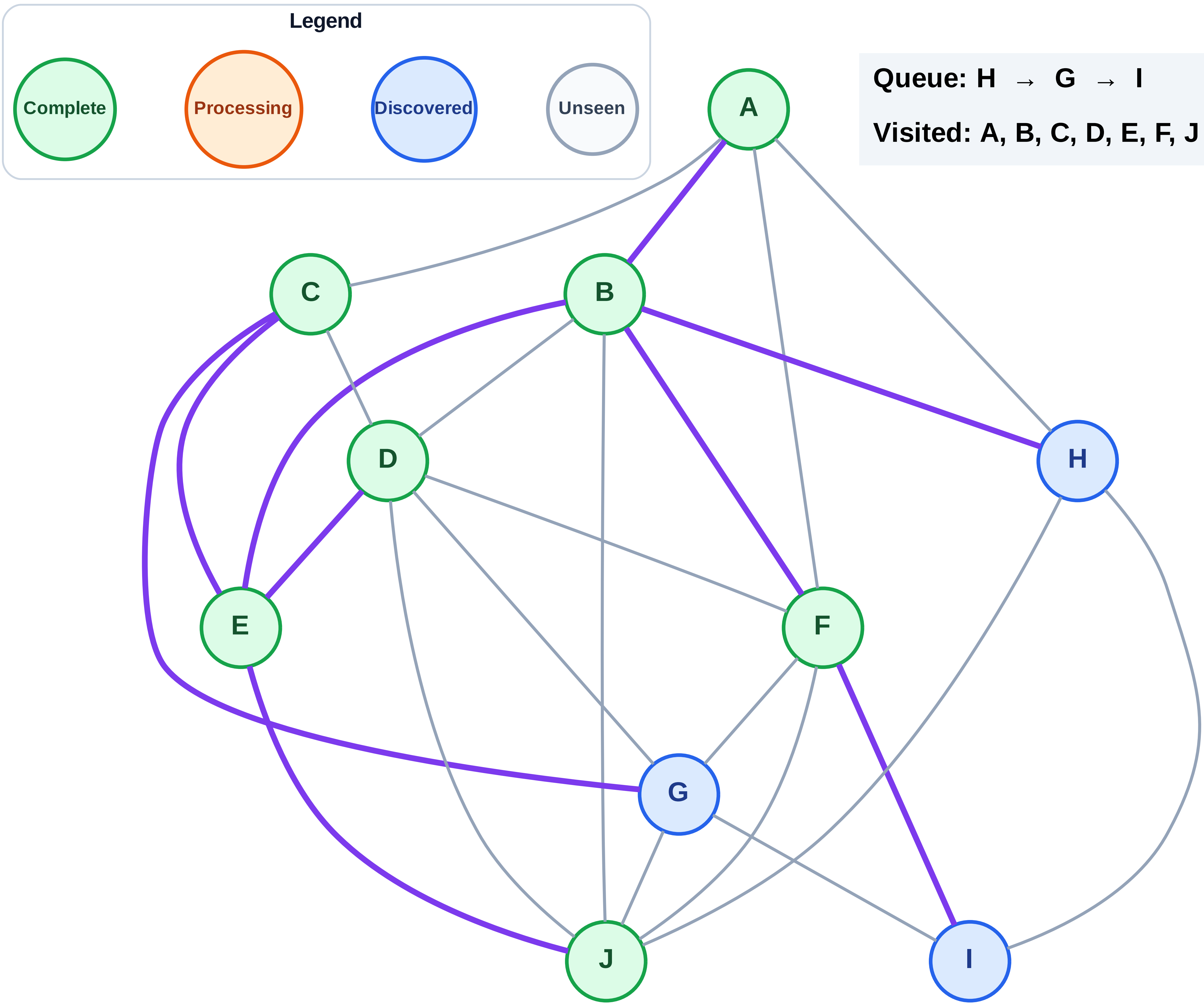
BFS Traversal

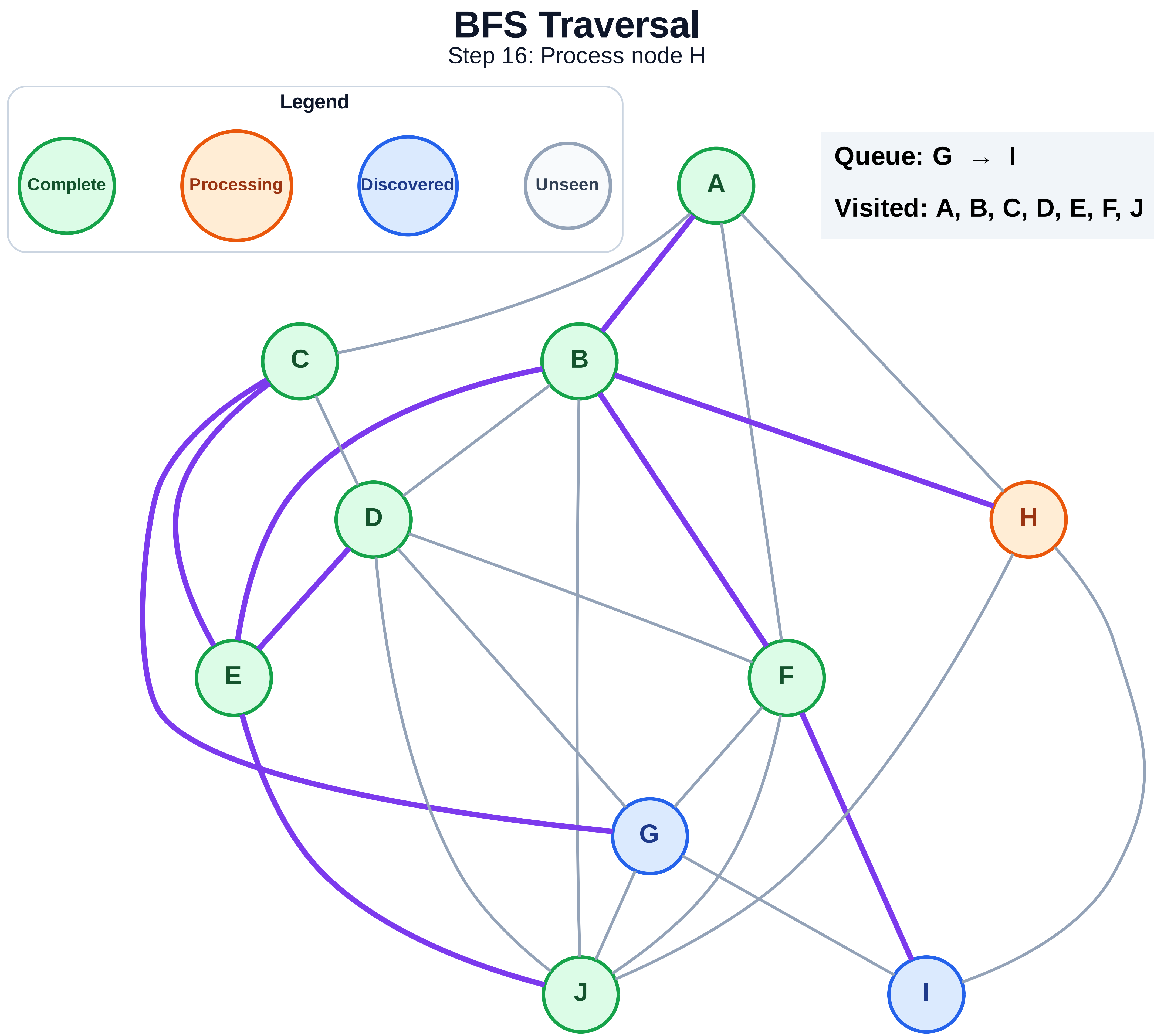
Step 14: Process node F



BFS Traversal

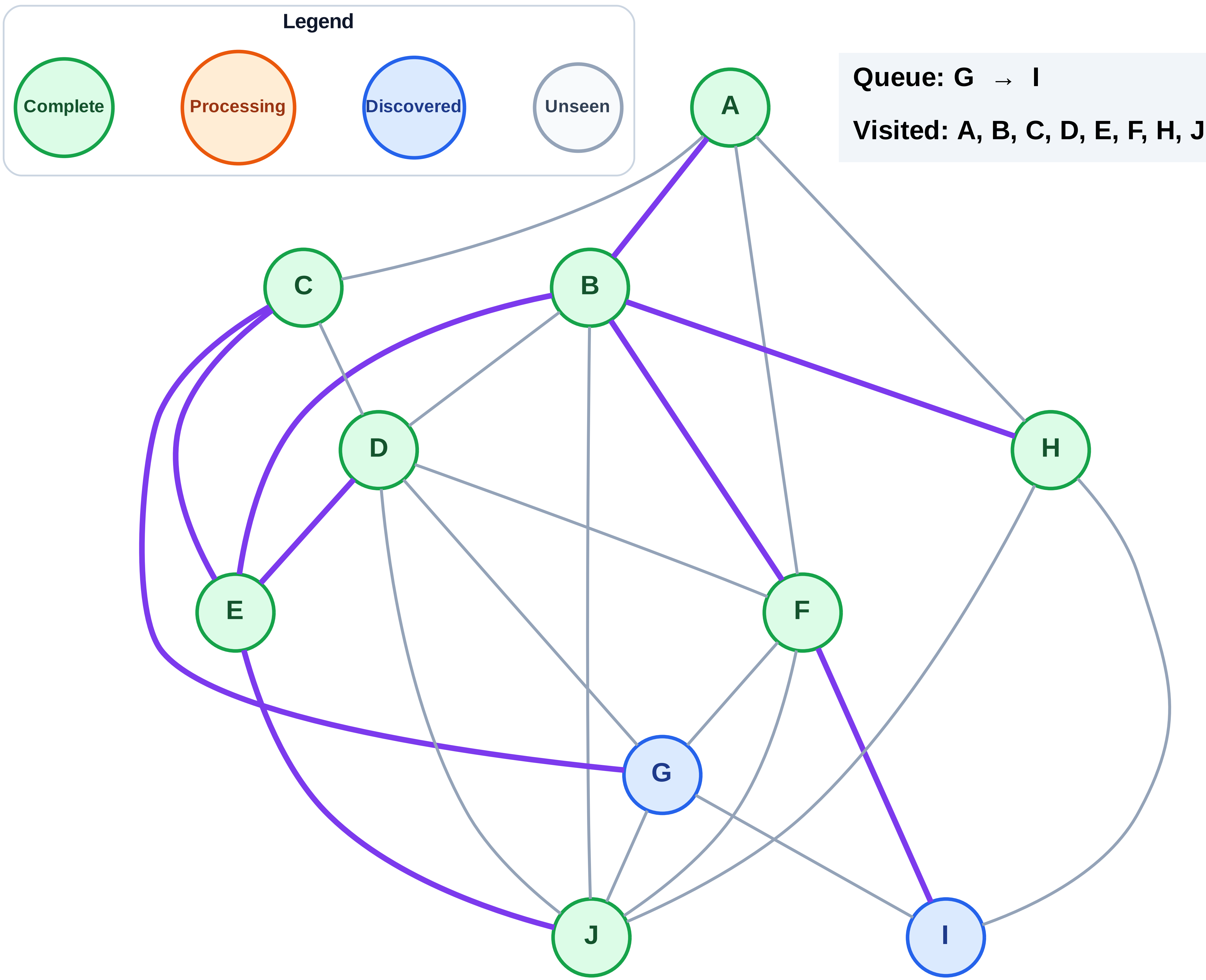
Step 15: Discover I from F




BFS Traversal
Step 16: Process node H
Legend
Complete
Processing
Discovered
Unseen
Queue: G → I
Visited: A, B, C, D, E, F, J
A
B
C
D
E
F
G
H
I
J

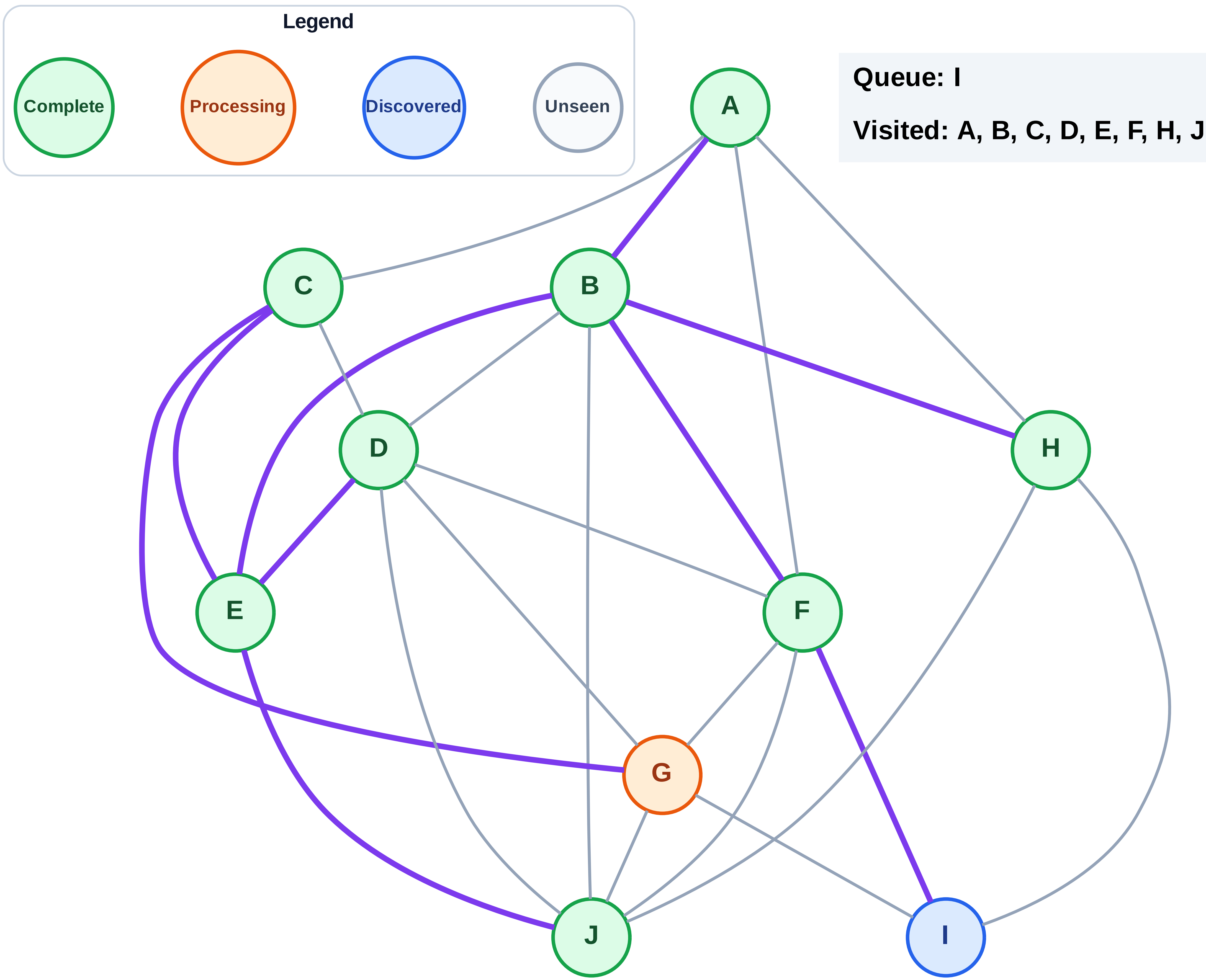
BFS Traversal

Step 17: No new neighbors from H



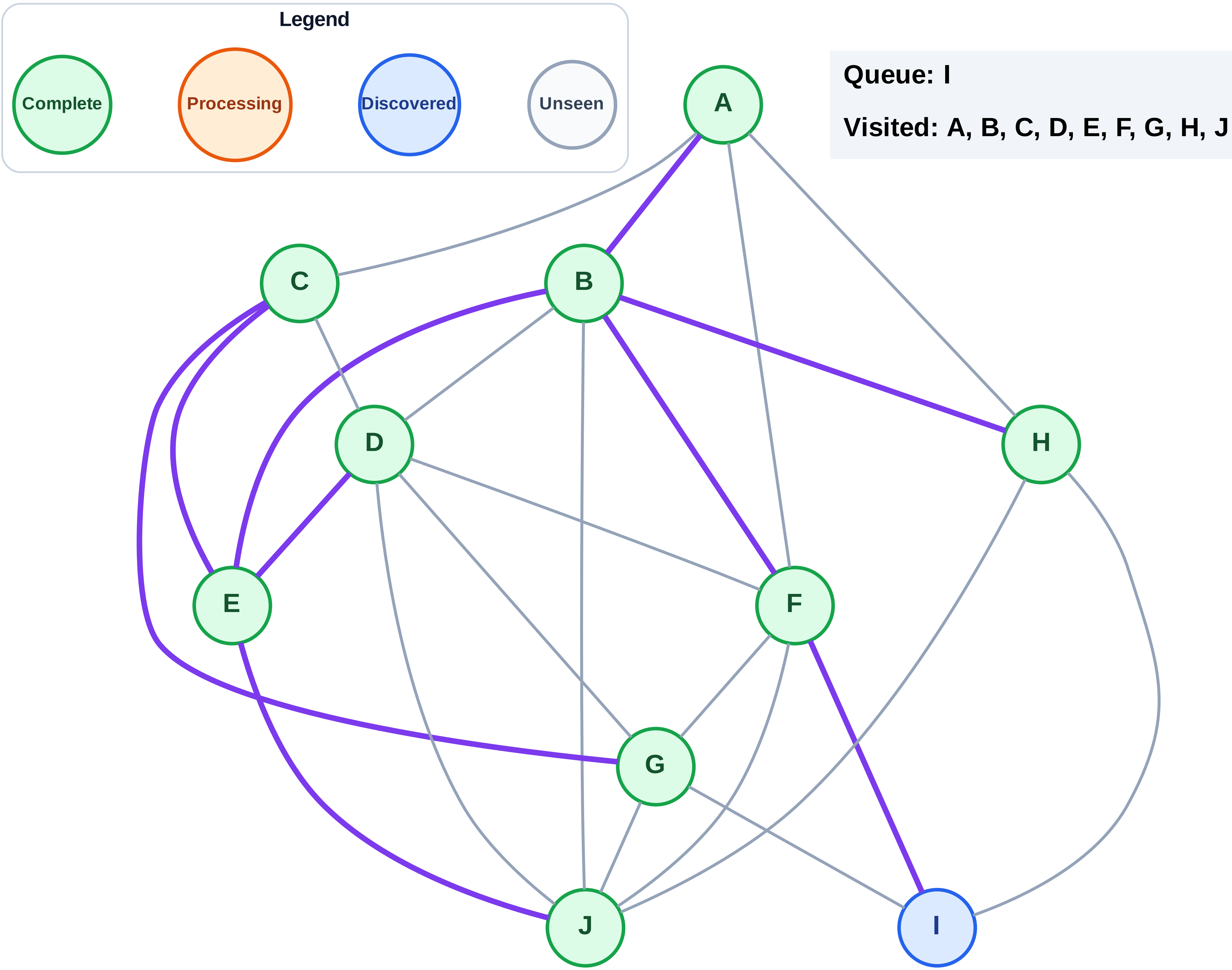
BFS Traversal

Step 18: Process node G



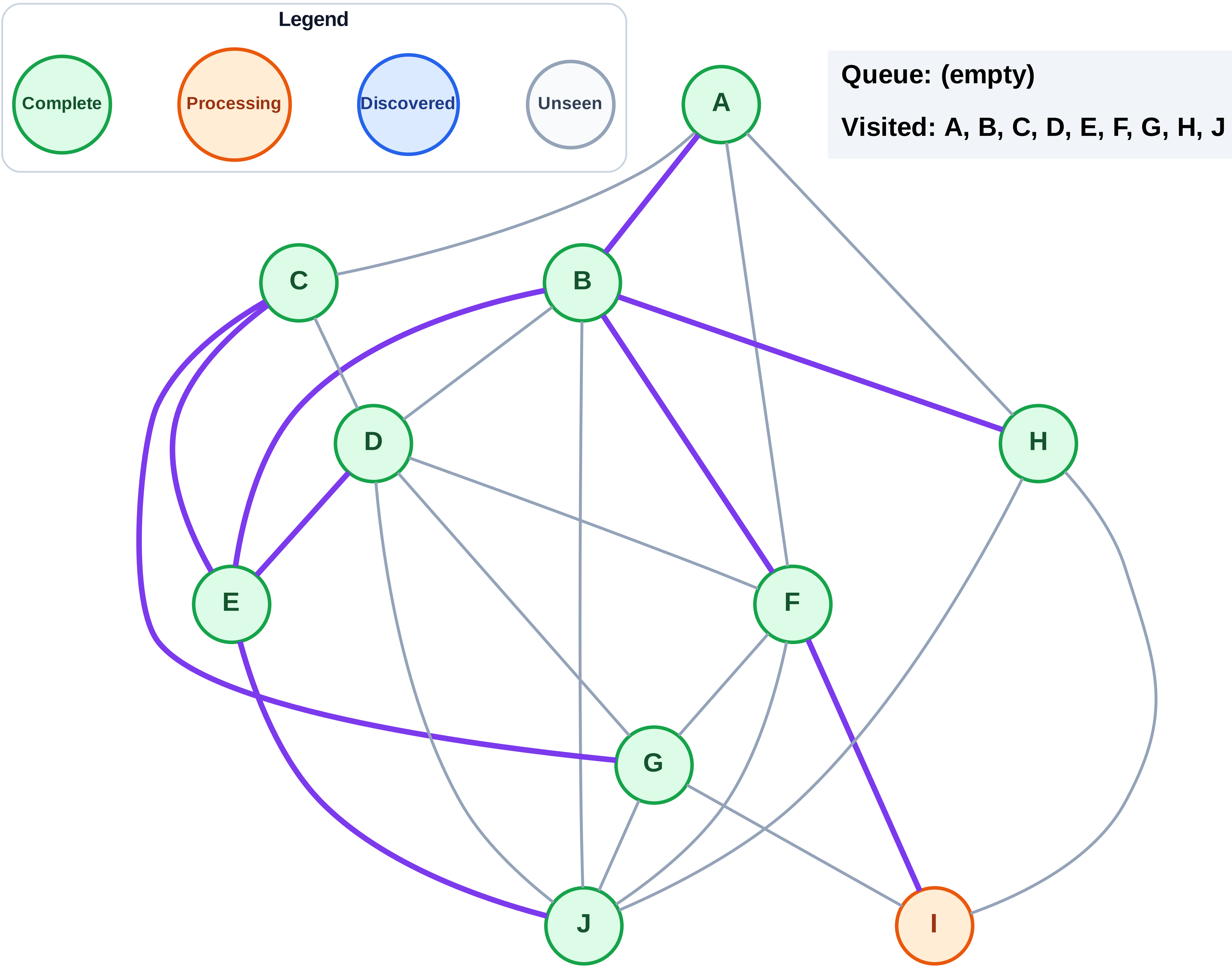
BFS Traversal

Step 19: No new neighbors from G



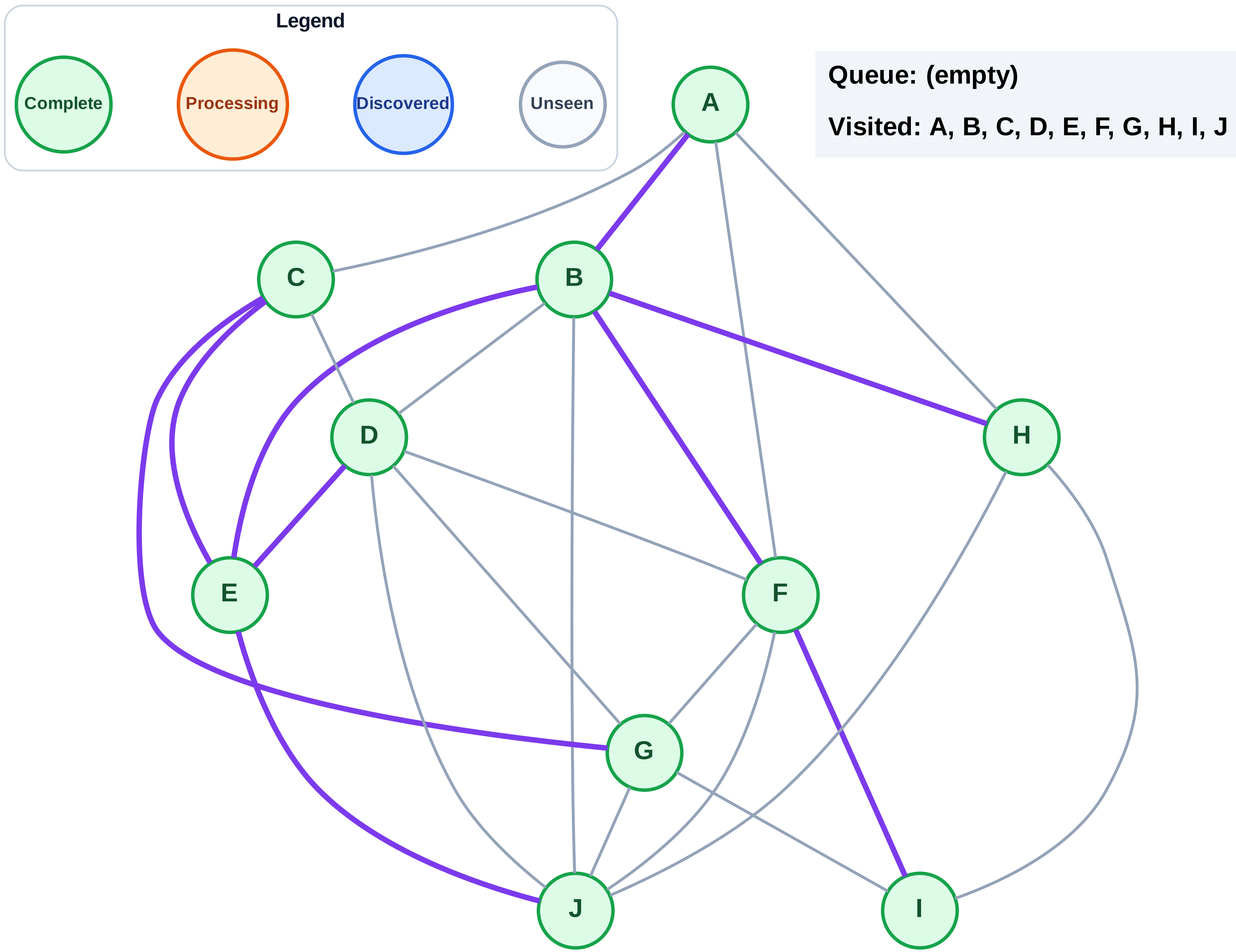
BFS Traversal

Step 20: Process node I



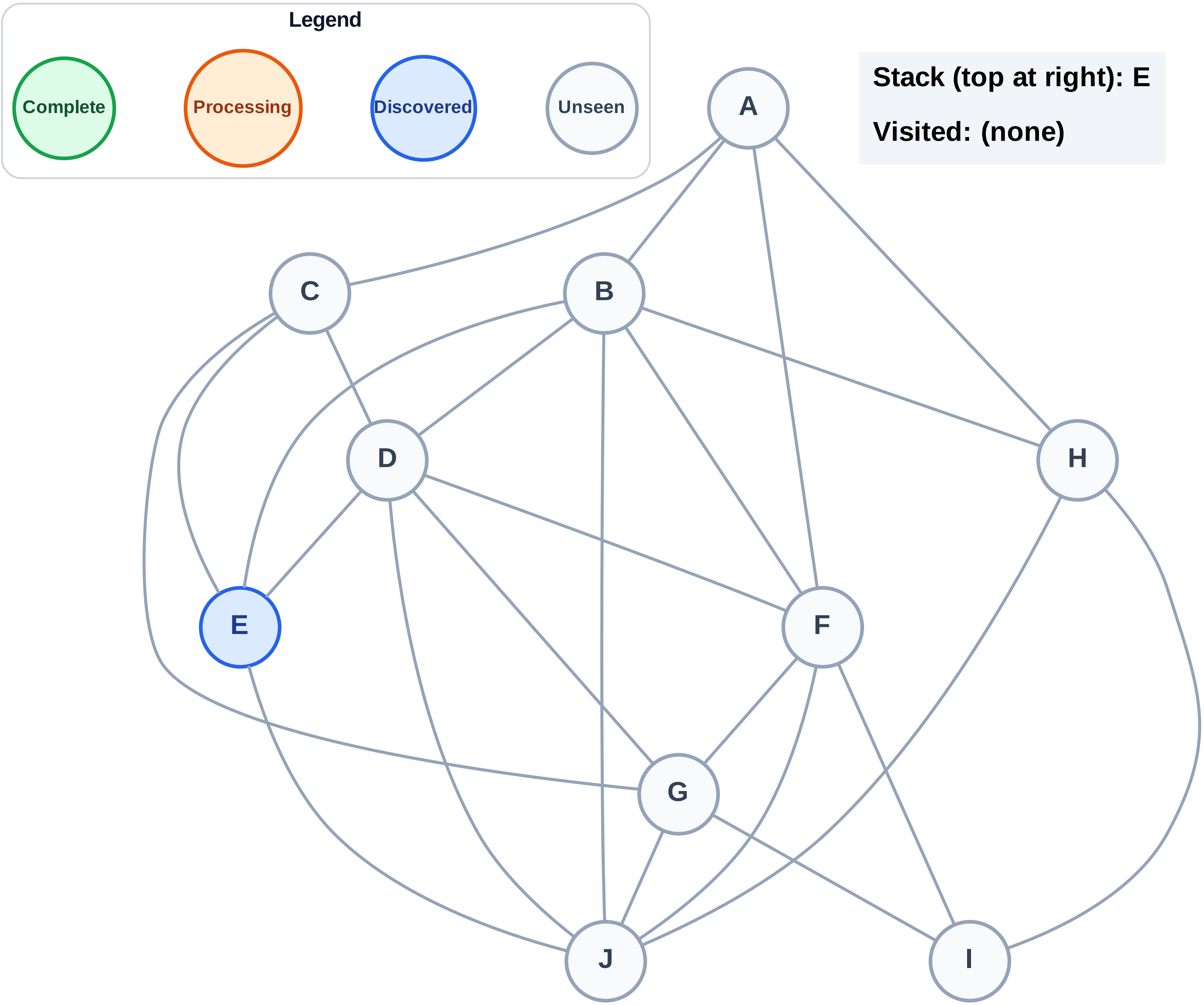
BFS Traversal

Step 21: No new neighbors from I



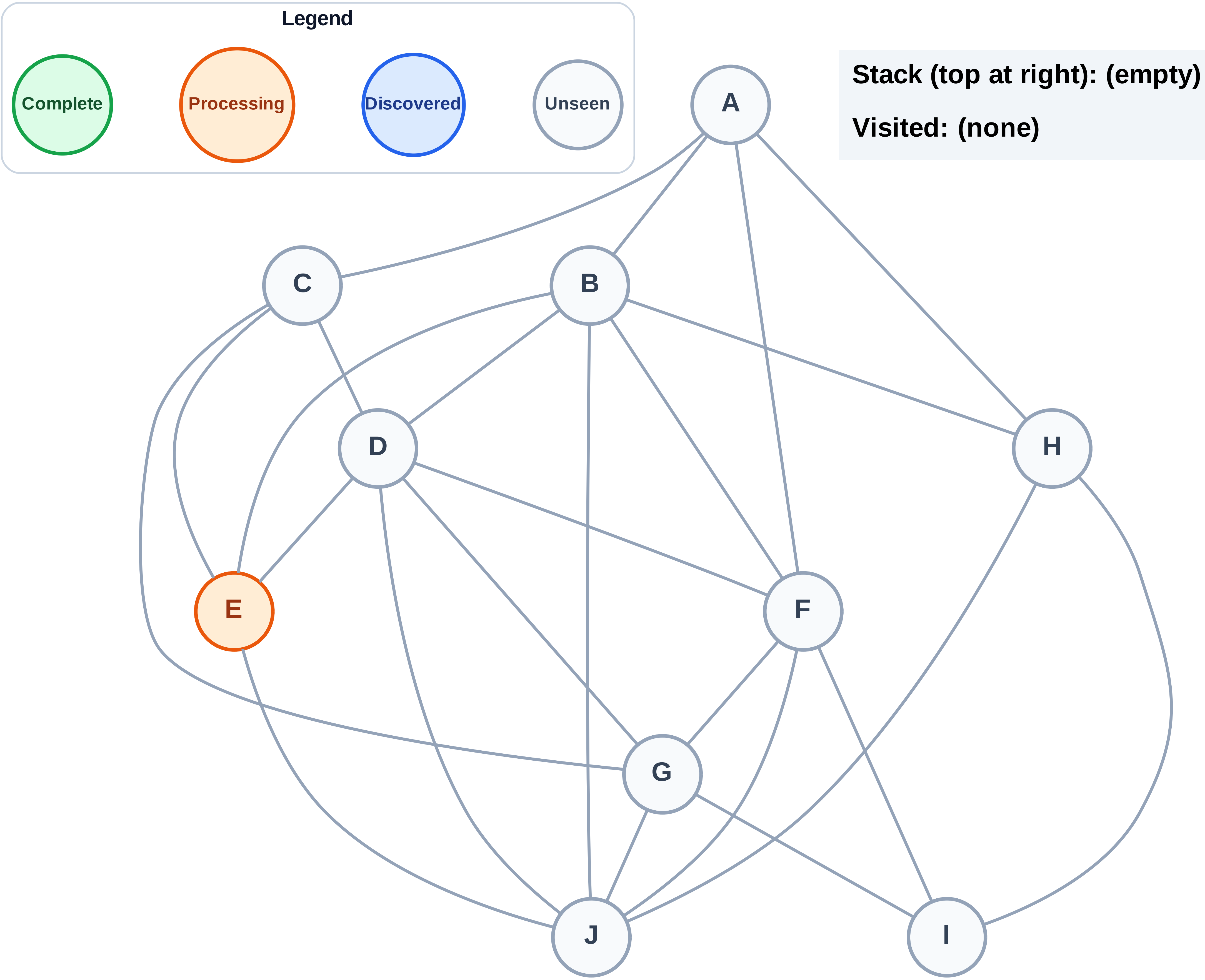
DFS Traversal

Step 1: Start at E



DFS Traversal

Step 2: Process node E

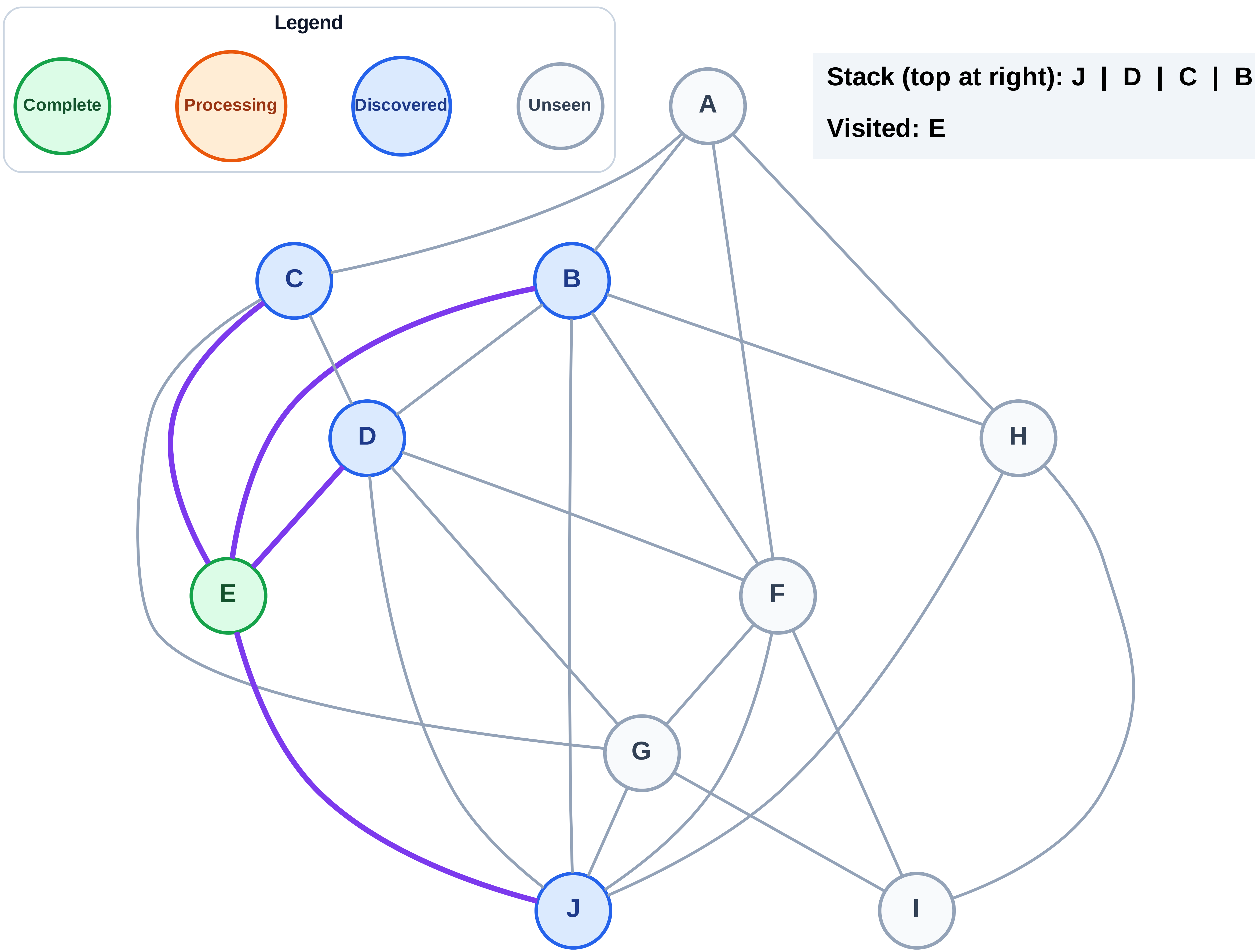


Stack (top at right): (empty)

Visited: (none)

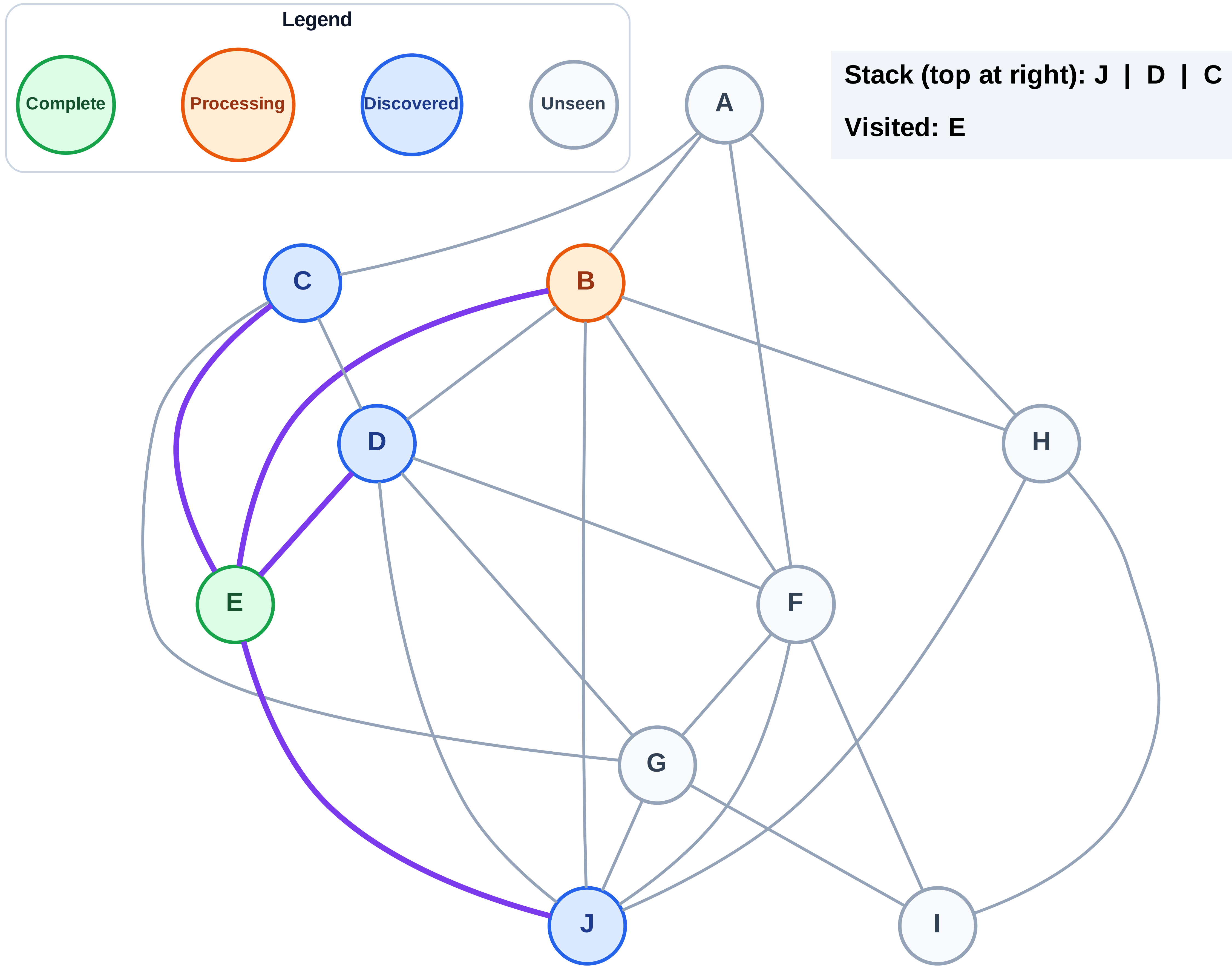
DFS Traversal

Step 3: Discover J, D, C, B from E



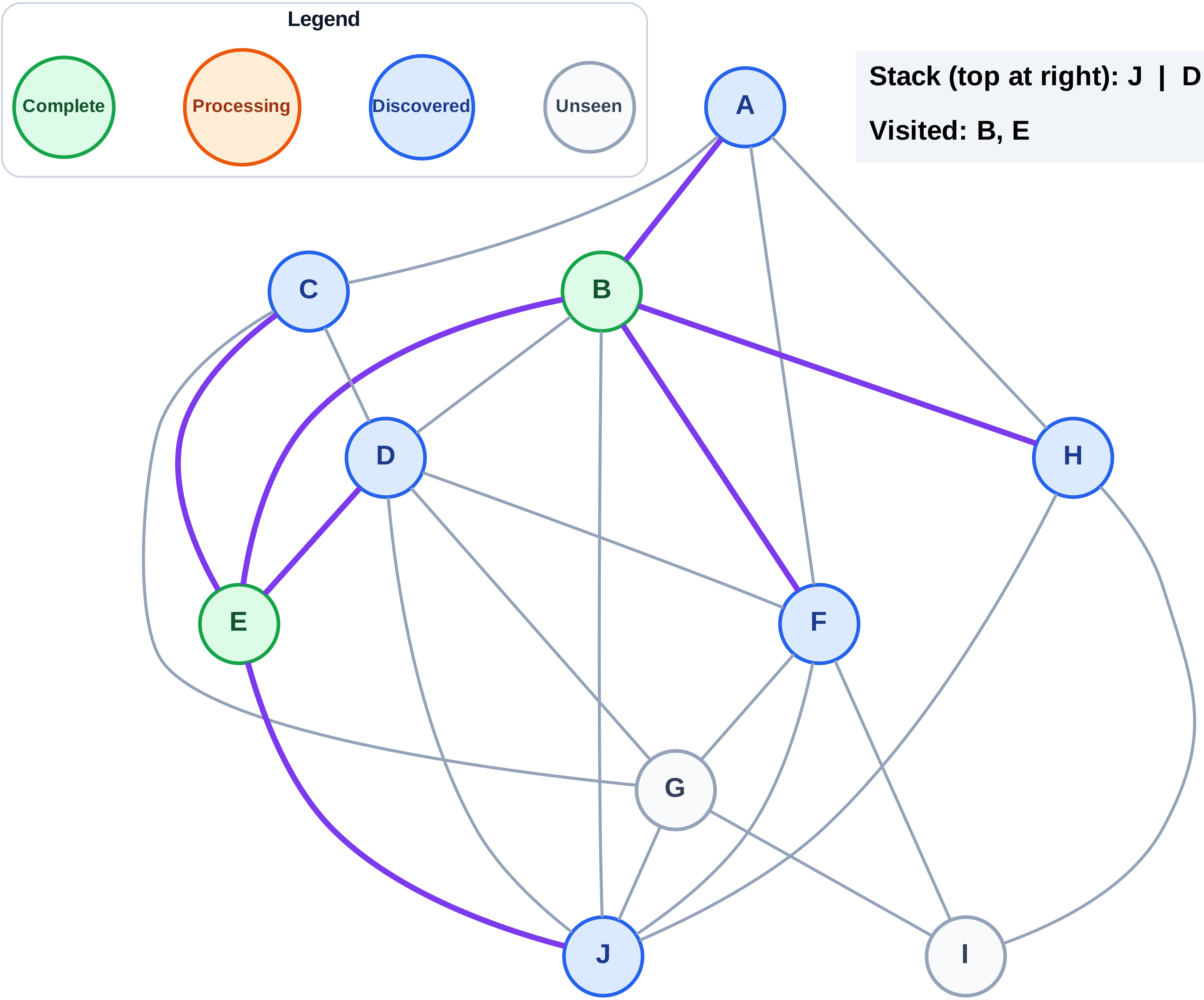
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover H, F, A from B

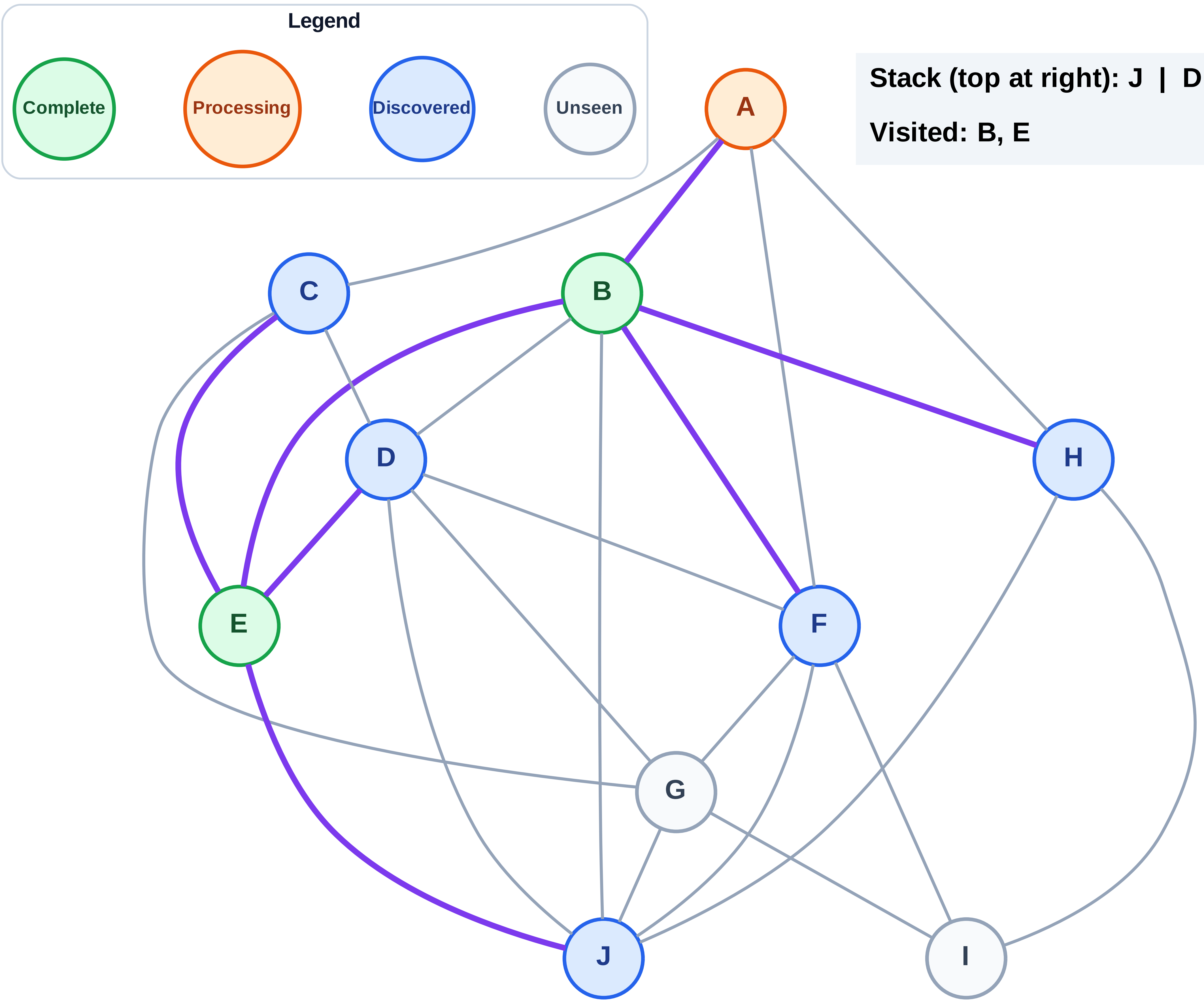


Stack (top at right): J | D | C | H | F | A

Visited: B, E

DFS Traversal

Step 6: Process node A

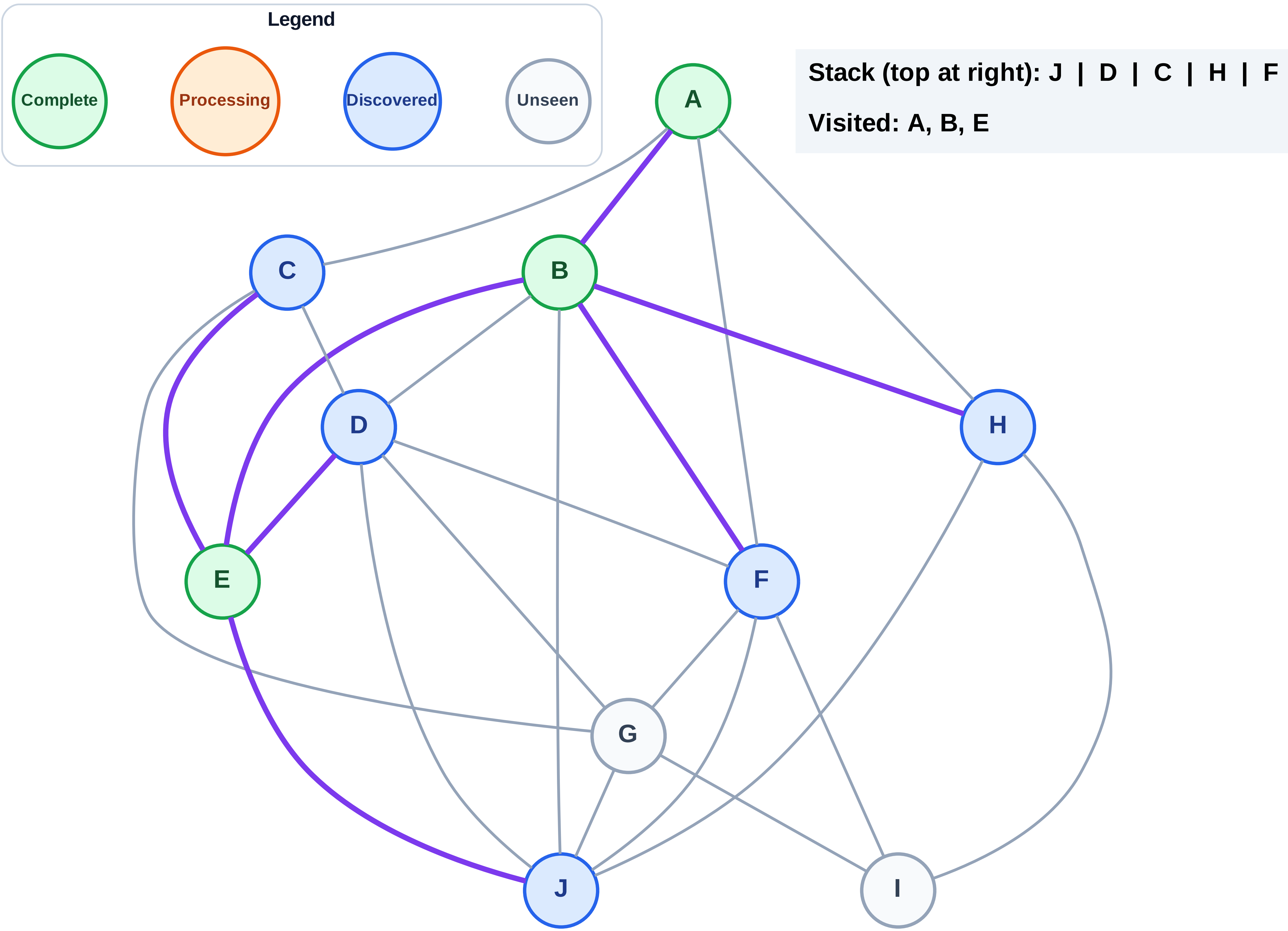


Stack (top at right): J | D | C | H | F

Visited: B, E

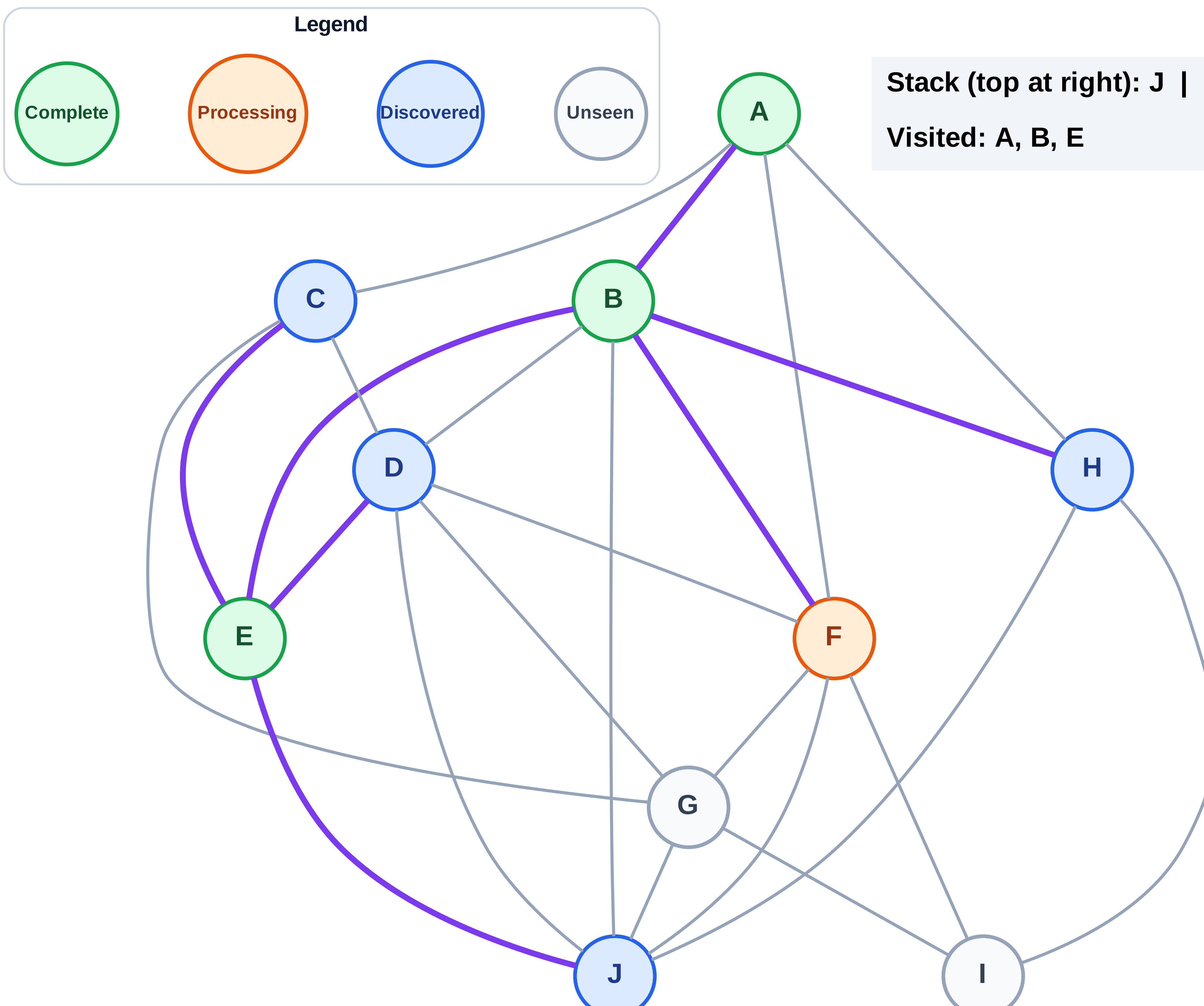
DFS Traversal

Step 7: No new neighbors from A



DFS Traversal

Step 8: Process node F

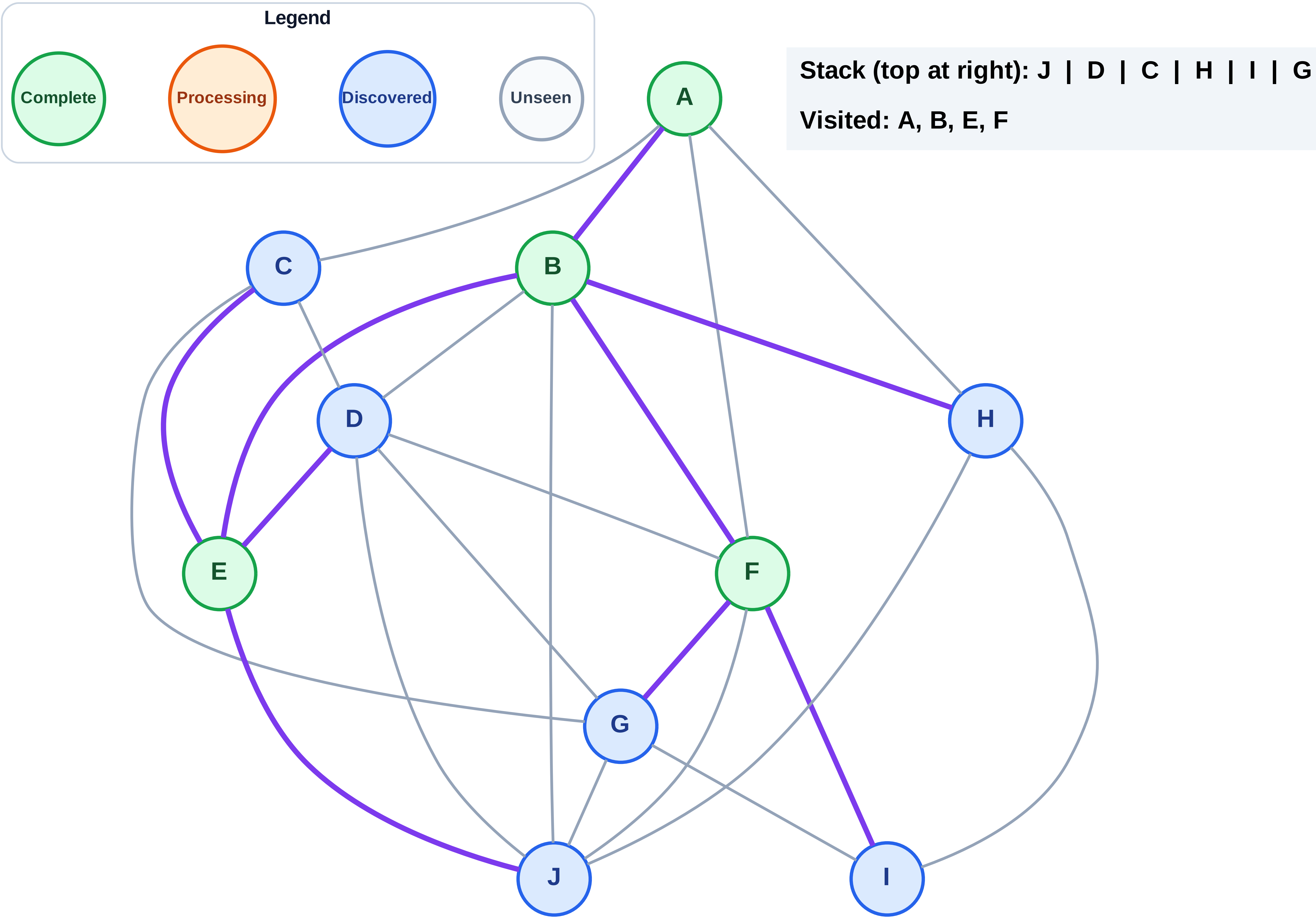


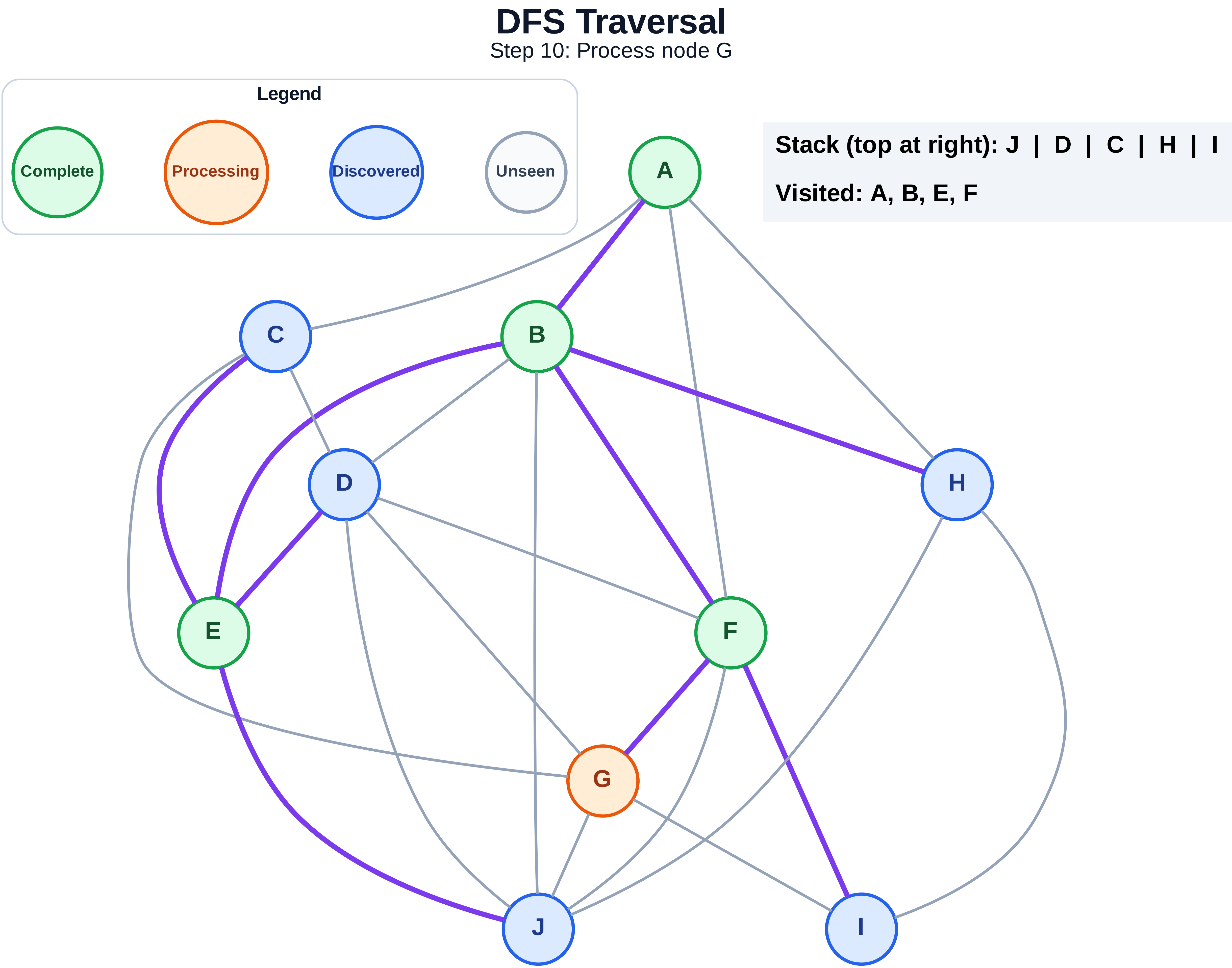
Stack (top at right): J | D | C | H

Visited: A, B, E

DFS Traversal

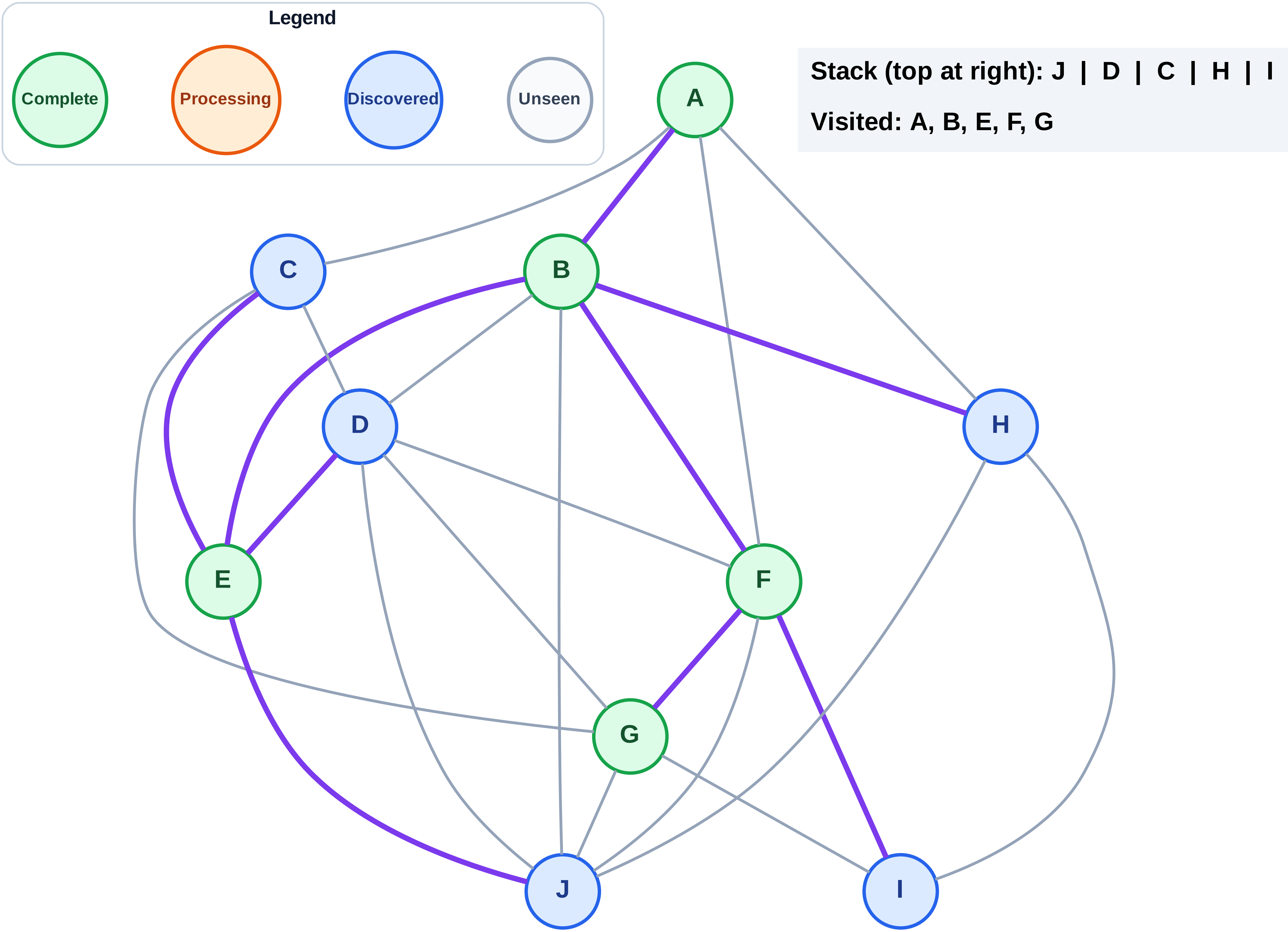
Step 9: Discover I, G from F

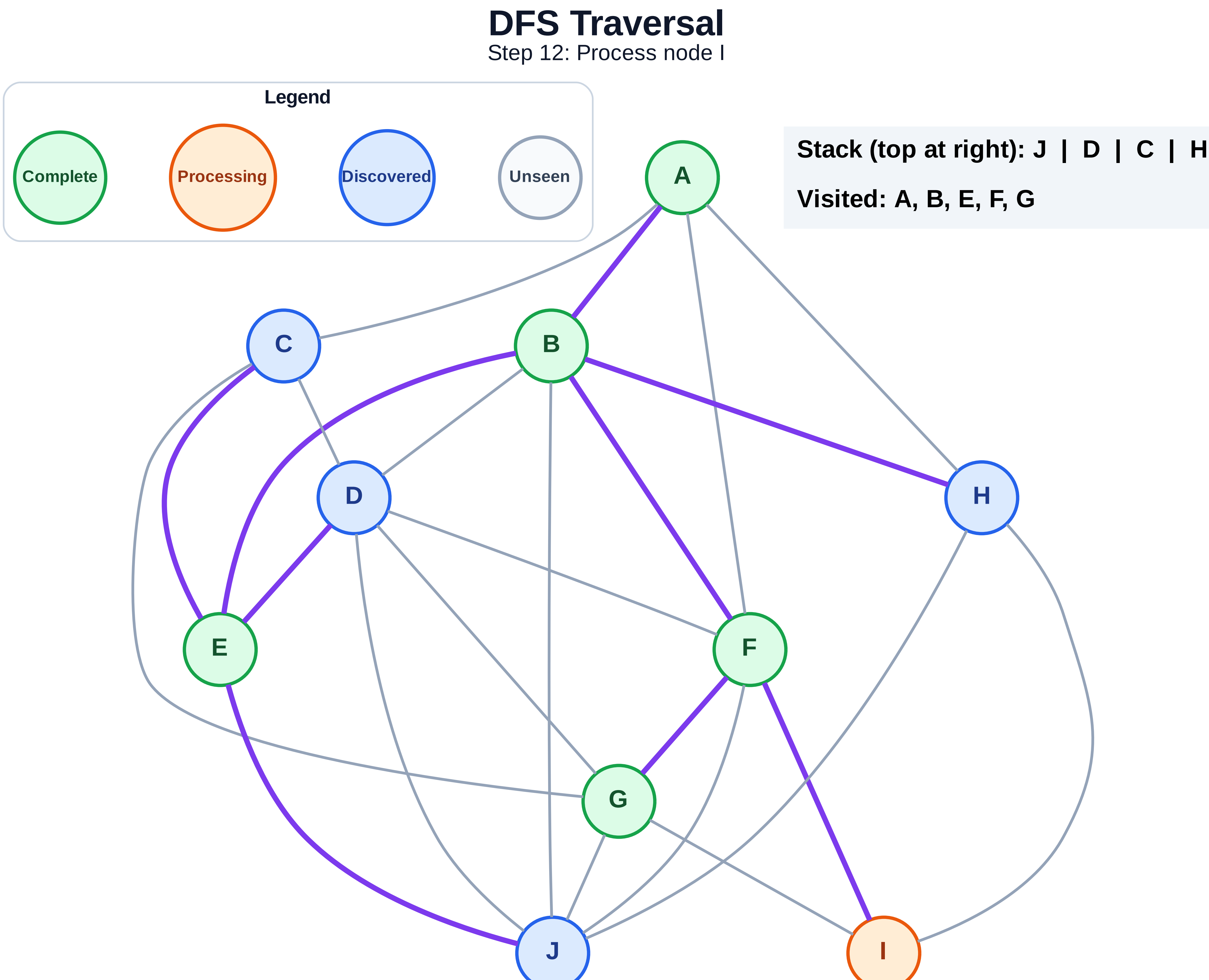



DFS Traversal
Step 10: Process node G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | D | C | H | I
Visited: A, B, E, F
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 11: No new neighbors from G

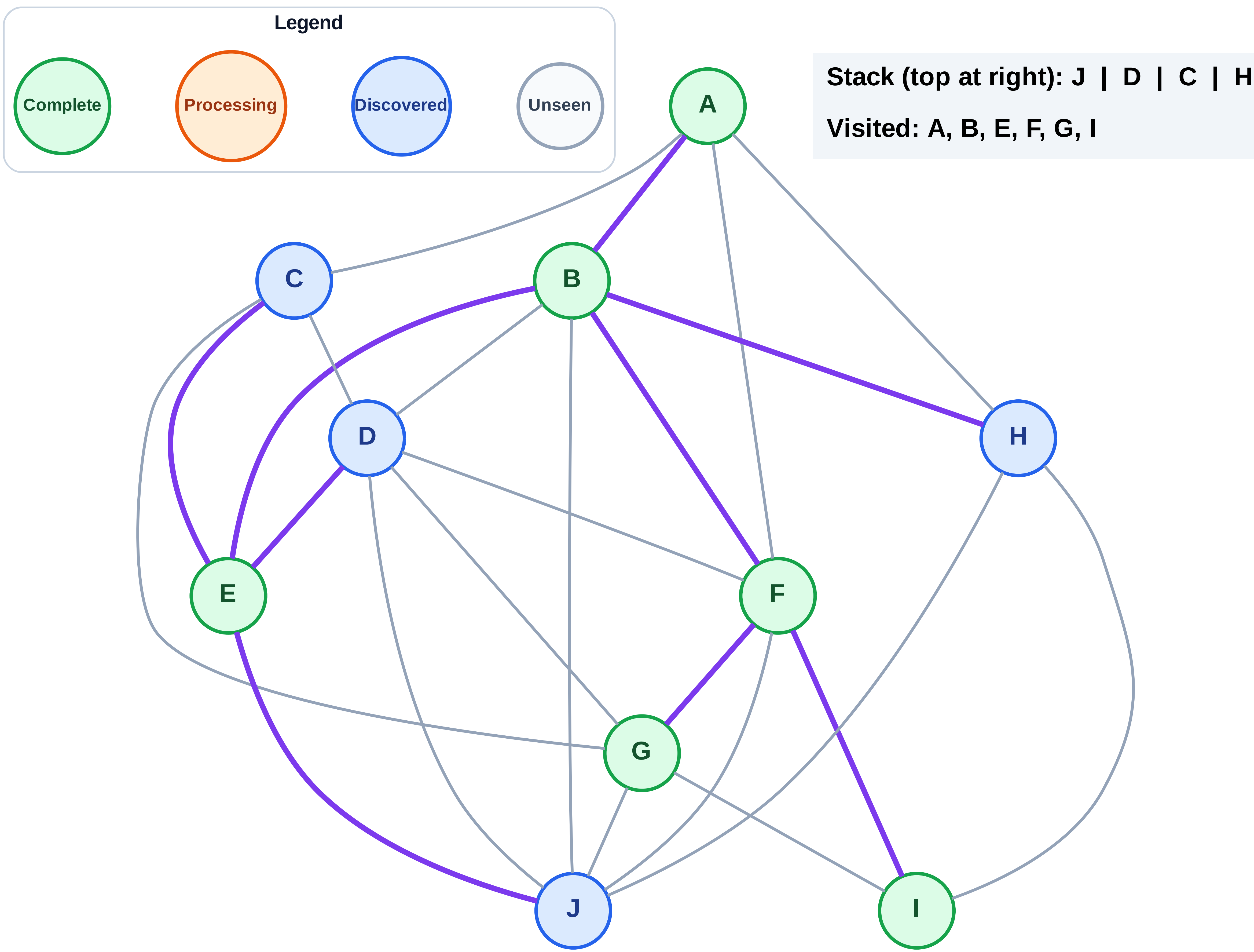




DFS Traversal
Step 12: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | D | C | H
Visited: A, B, E, F, G
A
B
C
D
E
F
G
H
I
J

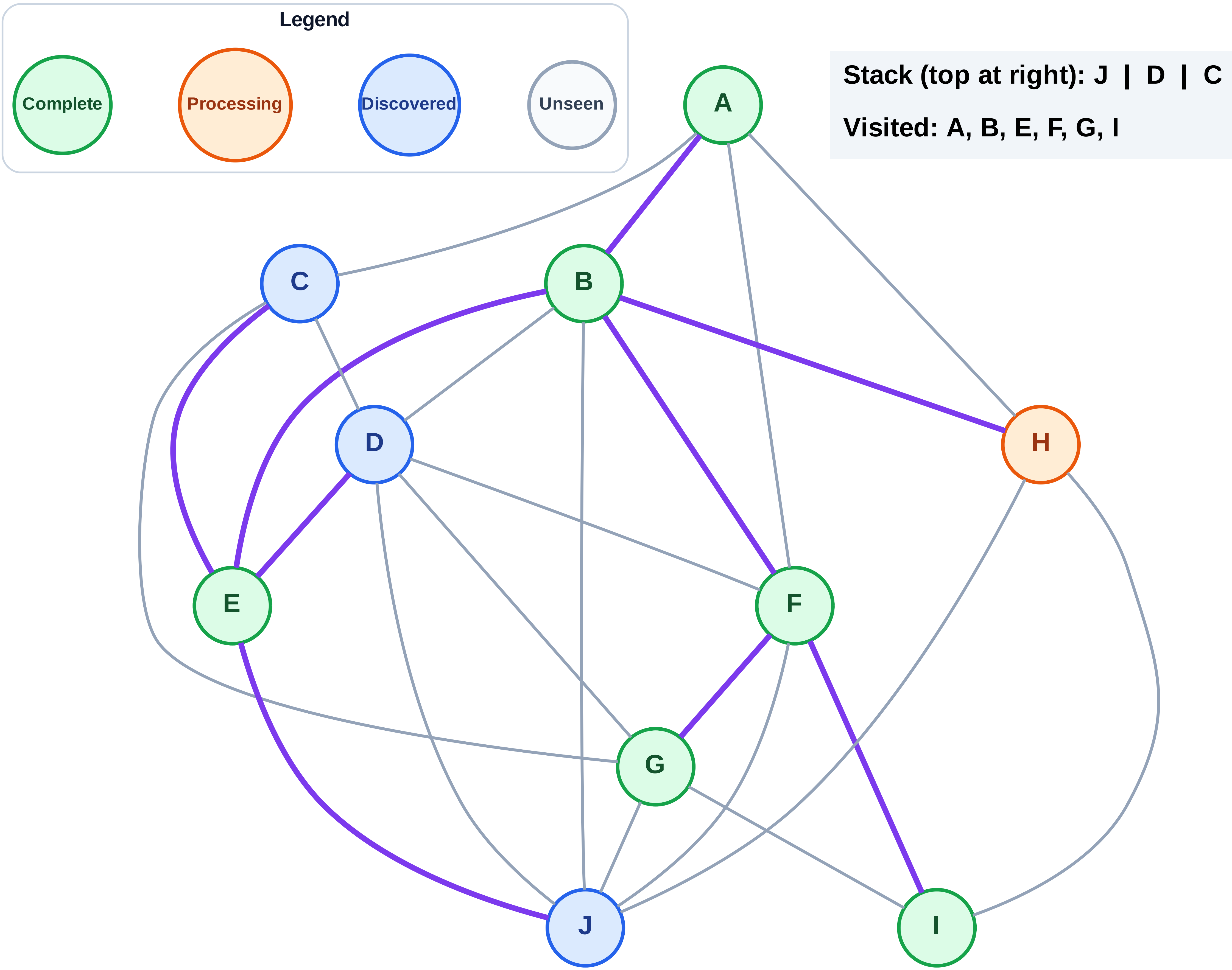
DFS Traversal

Step 13: No new neighbors from I



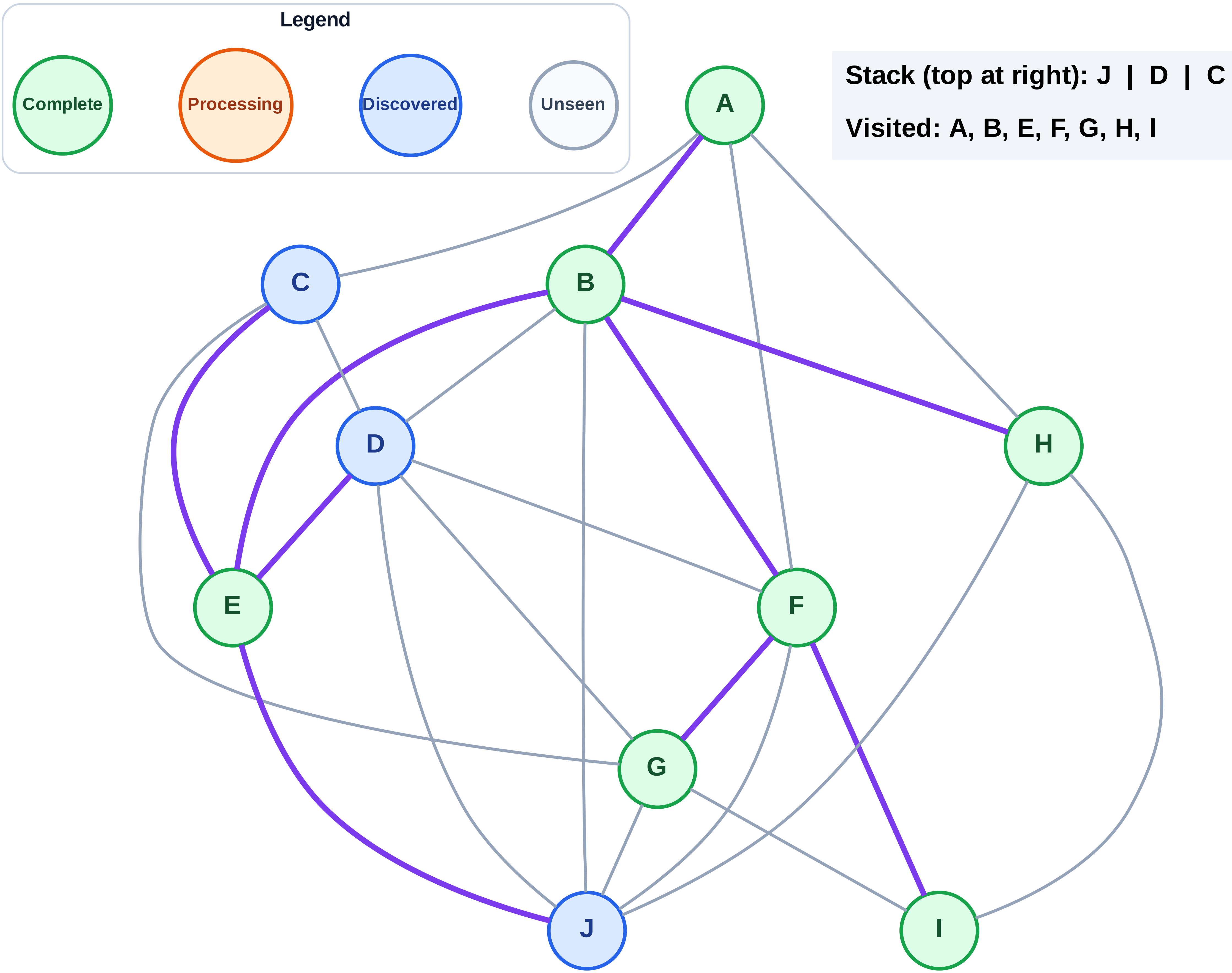
DFS Traversal

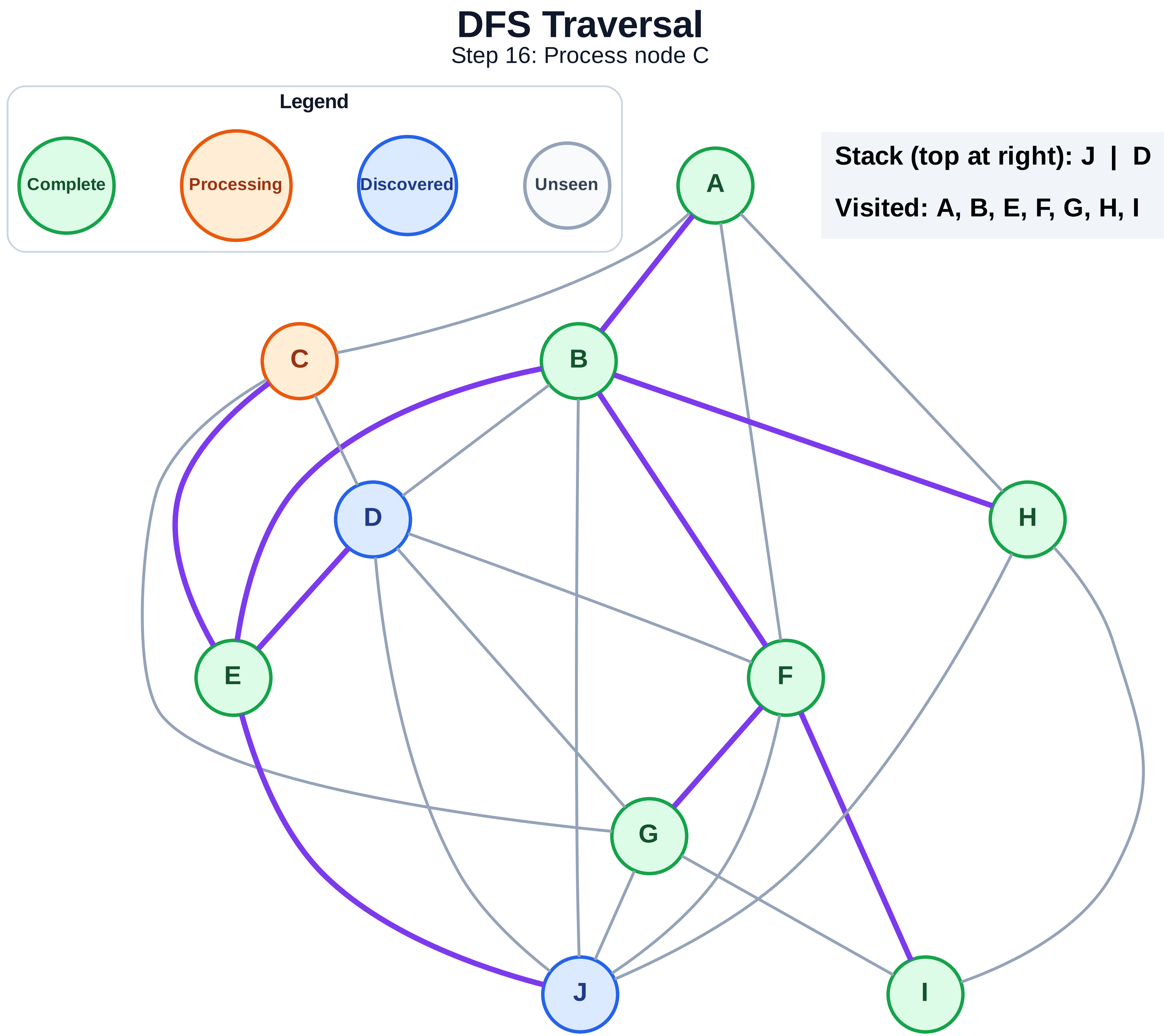
Step 14: Process node H



DFS Traversal

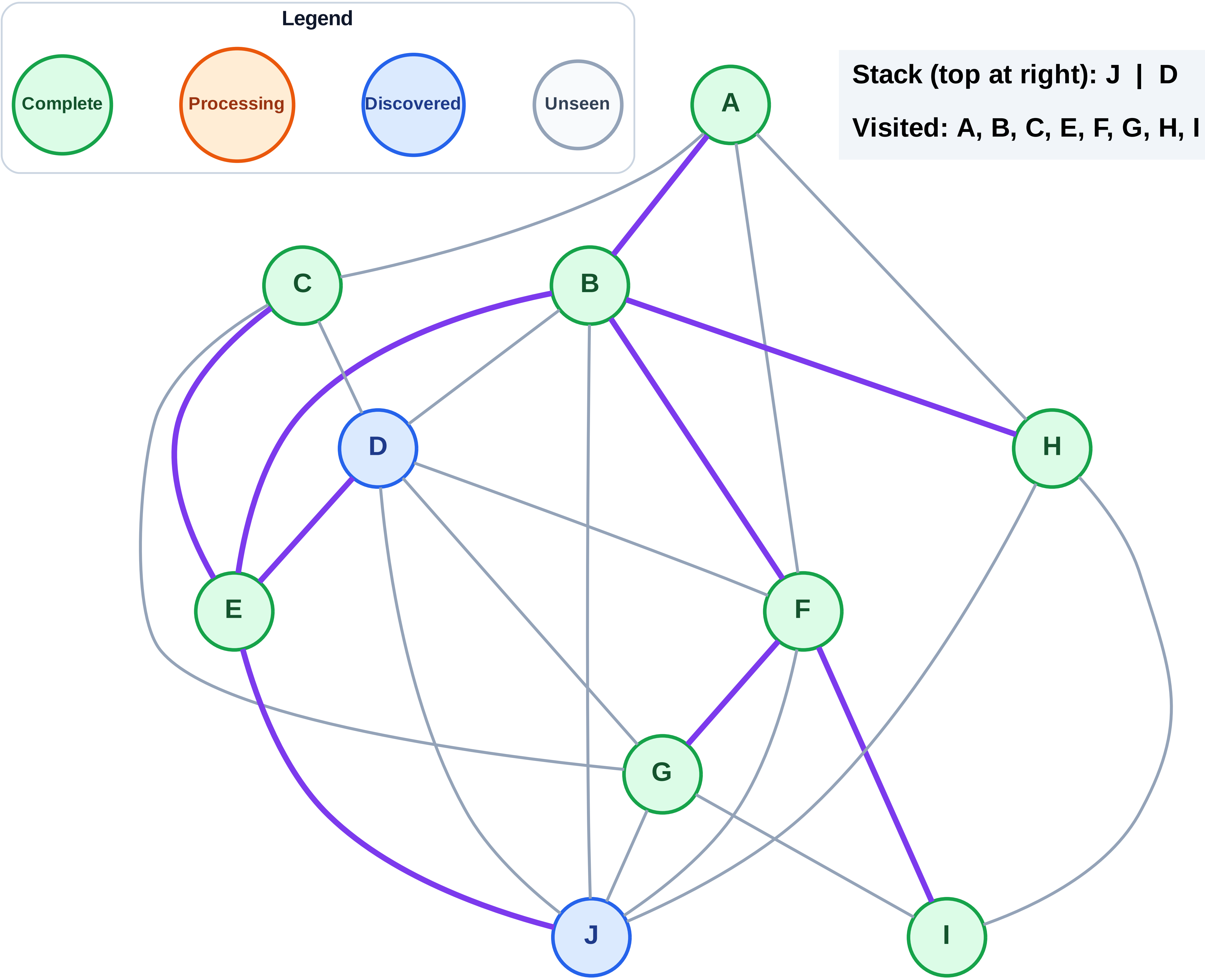
Step 15: No new neighbors from H




DFS Traversal
Step 16: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | D
Visited: A, B, E, F, G, H, I
A
B
C
D
E
F
G
H
I
J

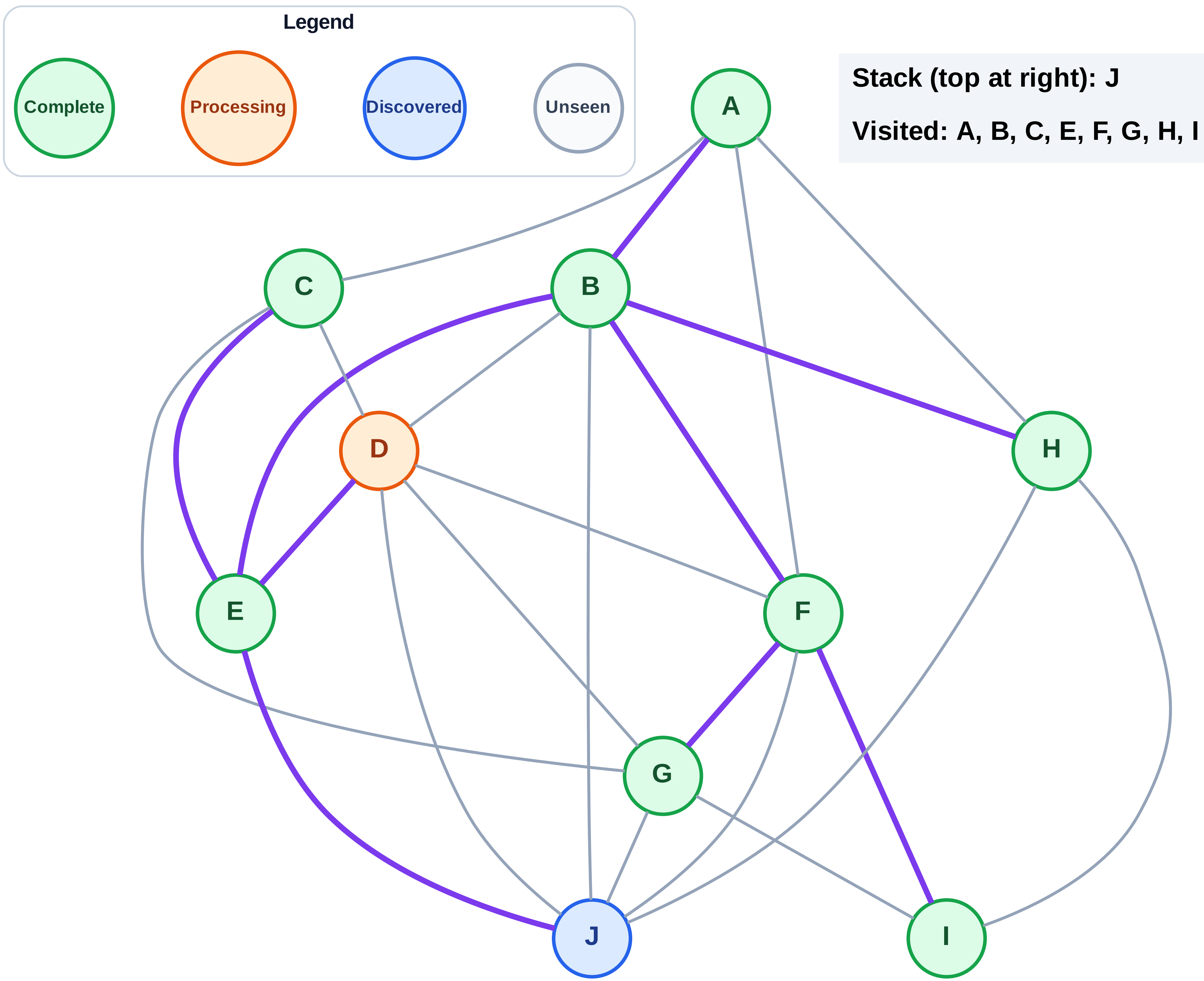
DFS Traversal

Step 17: No new neighbors from C



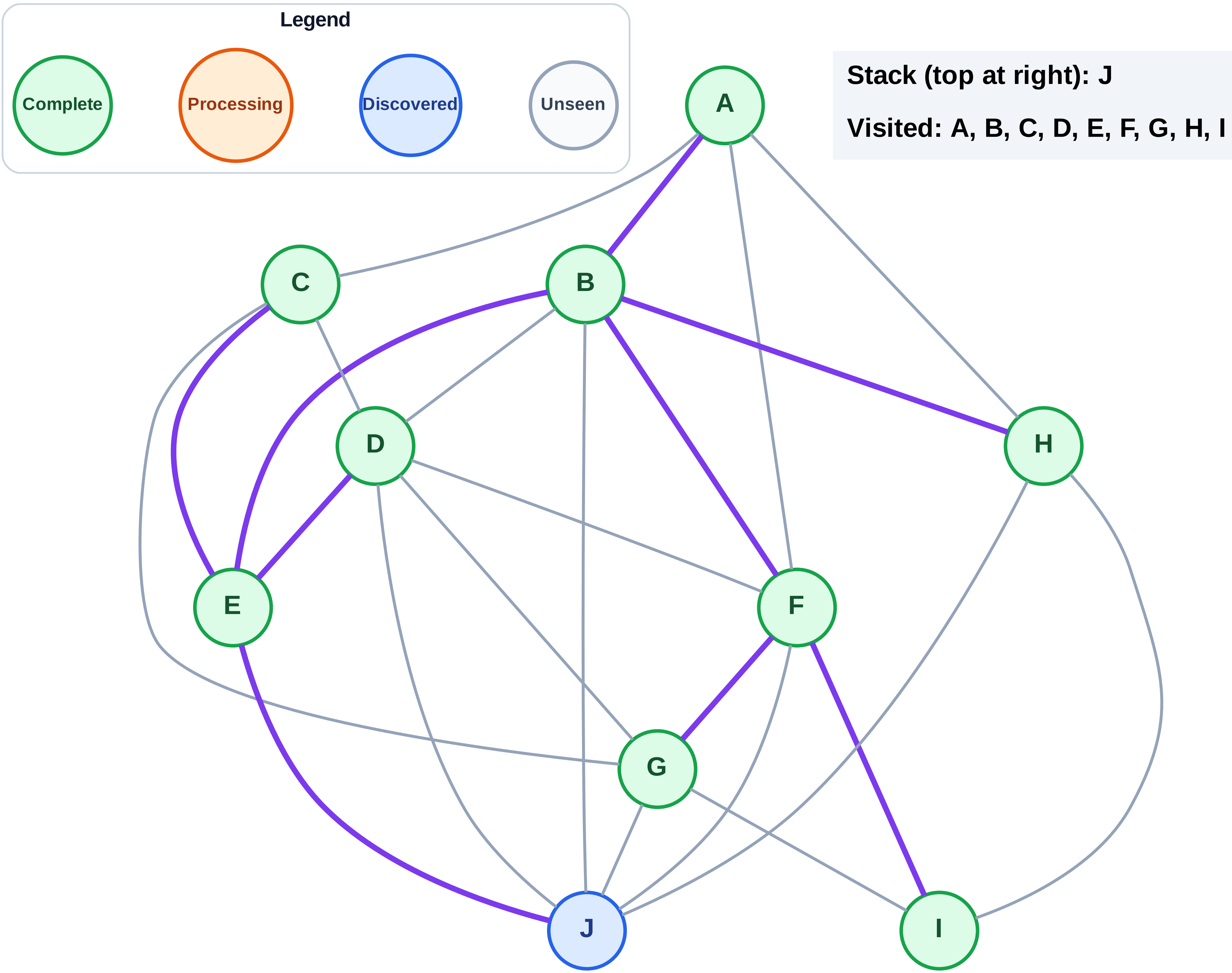
DFS Traversal

Step 18: Process node D



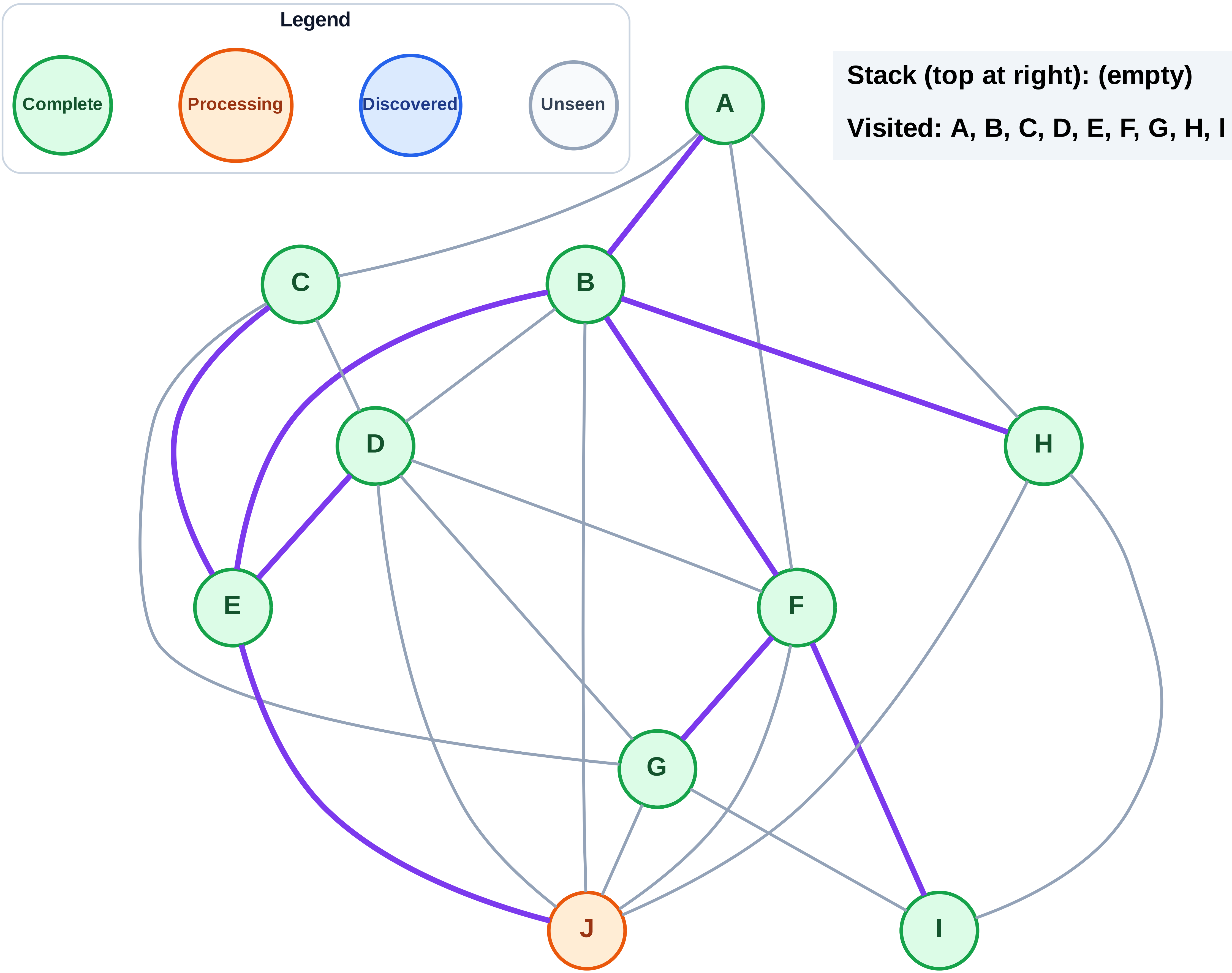
DFS Traversal

Step 19: No new neighbors from D



DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J